Software Design Document

MyHomeLib – Personal Library Cataloging App

Version 1.4

Author:
Grace Hechavarria

Peer Review Partner:
Yousef Habouh

Professor: Dr. Kiavash Bahreini

CEN 5064 Software Design – Fall 2026

Florida International University

October 10, 2026

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Grace Hechavarria	Initial release: architecture, components, detailed design, sprint planning, design rationale, and verification of the midterm build.	October 10, 2026
1.1	Grace Hechavarria	Sprint 7 hardening: ISBN check-digit validation, one error type and a timeout for unreachable book sources, recovery from damaged storage, books unique per owner in SQLite, one ISBN-13 key per book (ADR-010); scope for the final frozen to four core features, with the second-half plan aligned to it; 145 automated tests.	October 18, 2026
1.2	Grace Hechavarria	SOLID refactor (Interface Segregation): `BookRepository` no longer extends `IsbnCache`, so use cases depend only on the library methods they call and the planned sync store need not cache lookups; 146 automated tests.	November 6, 2026
1.3	Grace Hechavarria	Creational pattern map (`docs/pattern-map.md`); Abstract Factory for the device services (`DeviceFamily`: `PhoneDevices` / `BrowserDevices`, ADR-011); 151 automated tests.	November 6, 2026
1.4	Grace Hechavarria	Midterm feedback on the document: every diagram enlarged, split or laid out again to print at 7pt or more (it was 2.6–5.8pt); a one-page Key Design Decisions section up front; the proposed timeline condensed and repeated figures and text removed.	November 7, 2026
1.5	Grace Hechavarria	Core feature 4 built: email accounts and library sync through Supabase (ADR-012) instead of the planned Google sign-in; sign-in required to see	November 7, 2026

Contents

Key Design Decisions **4**

1 Introduction **6**
1.1 Purpose . . . 6
1.2 Customer, Setting, Sponsoring Organization, and Constraints . . . 6
1.3 Problem, Background, and Stakeholders . . . 7
1.4 Application Overview: Characteristics, Features, Differentiation, and Impact 8
1.5 Scope . . . 8
1.6 Definitions, Acronyms and Abbreviations . . . 9
1.7 Acknowledgements . . . 10
1.8 References . . . 10

2 System Overview **10**

3 System Components **13**
3.1 Decomposition Description . . . 13
3.2 Dependency Description . . . 16
3.3 Interface Description . . . 18
3.3.1 Presentation View Model . . . 18
3.3.2 Service Use Cases . . . 19
3.3.3 Domain Model . . . 20
3.3.4 Book Repositories . . . 20
3.3.5 Book-Source Adapters . . . 21
3.3.6 Accounts and Sync . . . 23
3.3.7 Device Adapters . . . 24
3.4 Module Interfaces . . . 24
3.5 User Interfaces (GUI) . . . 26
3.5.1 To Be Read . . . 26
3.5.2 Owned Catalog . . . 27
3.5.3 Wanted . . . 27
3.5.4 Scan . . . 28
3.5.5 Book Detail . . . 29

4 Detailed Design **29**
4.1 Data Detailed Design . . . 33
4.2 RTM . . . 36

5 Sprint Planning **38**
5.1 Role Focuses . . . 38
5.2 Sprint Schedule, User Stories, and Acceptance Criteria . . . 38
5.2.1 Proposed Timeline . . . 38
5.2.2 Actual Timeline . . . 39
5.3 Plan for the Second Half . . . 43

6 Design **46**
6.1 Definition of Design . . . 46
6.2 Design Process . . . 46
6.3 Graphical Representation of the Design . . . 46
6.3.1 Use Case Diagram . . . 46
6.3.2 Activity Diagram . . . 48
6.3.3 Class Diagram . . . 50
6.3.4 Sequence Diagram . . . 53
6.4 Rationale Management . . . 54
6.4.1 Issues Addressed . . . 54
6.4.2 Alternatives Considered . . . 55
6.4.3 Decisions Made . . . 55
6.4.4 Criteria Used to Guide Decisions . . . 56
6.4.5 Design Debate and Final Resolution . . . 56

7 Verification **56**
7.1 Sprint 1 – Project Proposal and Approval . . . 58
7.2 Sprint 2 – Tier Breakdown . . . 58
7.3 Sprint 3 – C4 and UML Models . . . 58
7.4 Sprint 4 – Scanning, Shelves, and Copies . . . 59
7.5 Sprint 5 – Acquisition Rules . . . 60
7.6 Sprint 6 – Exact Edition, Refactor, and Design System . . . 61
7.7 Sprint 7 – Hardening: Bad Input and Failures . . . 66
7.8 Sprint 8 – Accounts and Library Sync . . . 67

8 References **68**

Key Design Decisions

MyHomeLib is a phone app for cataloging a home library by ISBN. The decisions that shape it, each recorded as an Architecture Decision Record (ADR) in `docs/adr/`:

Decision	**What and why**	**ADR**
Four-tier modular monolith	One TypeScript codebase in four folders: Presentation → Service → Domain; Data implements the Service's ports and is wired in only by `bootstrap.ts`. The rules can be tested without a phone, and storage or book sources can change without touching them. Automated tests enforce the dependency rule.	001, 009
Expo Go on the phone	The review partner can run it without a Mac or a paid account. The browser build is a limited fallback (no camera).	005
SQLite on the device	One data store; the browser build keeps the same rows as JSON behind the same port, translated only by `BookRowMapper`.	002
Official book APIs, exact edition	Open Library, plus optional Google Books, asked together, merged edition-first and cached; no scraping. The owner confirms the cover that matches their copy.	003, 004, 006
Rules live in the domain	`Book`, `PersonalLibrary`, `Purchase` and `Gift` enforce the rules (a duplicate ISBN adds a copy; an Owned book needs a purchase or a named gift), so no screen can bypass them.	001
MVVM and a design system	One view-model class holds every screen's state and flow; components only render, with tokens and imported stylesheets.	007, 008
Device services as one family	The camera and haptic feedback come from one family per platform (Abstract Factory), chosen once.	011
Validate at the edges	A mistyped ISBN is refused before any network call; unreachable sources time out with one clear error; damaged saved data still loads.	010
Accounts and sync through Supabase	Email sign-in that runs in Expo Go (Google sign-in needs a native build that is not free on iPhone). The phone's store stays the working copy; a Supabase table holds each account's synced copy behind row level security. Per book the latest change wins, deletes travel as tombstones, and Realtime tells the other phones.	012

Scope for the final, frozen at the midterm check-in: (1) scan a book, (2) shelve it with the rules enforced, (3) browse and manage the library, and (4) account sign-in and library

sync. All four are built and tested; feature 4 was built with email accounts instead of Google sign-in (ADR-012).

1 Introduction

1.1 Purpose

What: MyHomeLib is a cross-platform mobile app (Expo Go on iPhone and Android, with a web fallback) for cataloging a personal home library by scanning each book's ISBN. **For whom:** a bibliophile who owns a growing collection of physical books and wants to know exactly what is on their shelves, what they still want, and what they have not read yet.

Core features (the scope for the final, frozen at the midterm check-in: nothing new is added until these four are complete and tested):

UC1 Scan a book: scan or type an ISBN to look up the exact edition from Open Library and Google Books, then pick the cover that matches your copy or photograph it.

UC2 Shelve it, with rules enforced: file the book as Owned or Wanted, and as read or unread. An Owned book must be marked purchased or gifted, and a gift needs the giver's name. Scanning a book you already have adds a copy instead of a duplicate.

UC3 Browse and manage your library: see To Be Read (unread and read), the Owned catalog grouped A–Z, by author or by series, and the Wanted list. From a book's page you can mark it read, move it, change its cover, or delete it.

UC4 Account sign-in and library sync: sign in with an email and a password (required to see or add books) to keep your library backed up and in sync on all your phones, automatically. Each phone's copy works offline; the latest change to a book wins, and a book deleted on one phone is deleted on the others.

Within course limits: one data store (SQLite on the device), a four-tier architecture, official book APIs only (no scraping), and an app the review partner can run without a Mac or paid accounts. This document describes that design for the course instructor, the peer review partner, and any engineer who will maintain the system: how it is divided into tiers and components, how they communicate, why the main decisions were made, and how each requirement was verified.

1.2 Customer, Setting, Sponsoring Organization, and Constraints

The primary customer is the end user: a bibliophile who owns a large or growing collection of physical books and wants to know exactly what is on their shelves, what they still want to buy, and what they have not read yet. The formal setting is the semester project for CEN 5064 Software Design at Florida International University. There is no external sponsor; the project is overseen by the course instructor, Dr. Kiavash Bahreini, and reviewed at two structured peer review sessions (midterm and final).

The course and instructor set the following constraints, which shaped the design:

- **Platform a partner can run.** The review partner must be able to clone the repository and run the app without a Mac, Xcode, Android Studio, or a paid developer account. This led to Expo Go (ADR-001, ADR-005).

- **One data store.** The project may use a single persistence technology. On a phone that is SQLite; on the web fallback the same port is backed by browser storage (ADR-002). Feature 4 keeps the device store as the only working store: the Supabase table is a synced copy of it, reached through separate ports (`SyncStore`, `LibraryBackup`), and ADR-012 records it as a second store.
- **Limited external integrations, and no scraping.** Bookstore web scraping was proposed early and dropped at the instructor's request (ADR-004). Book metadata comes only from official APIs: Open Library (no key) and, optionally, Google Books (free key) (ADR-003, ADR-006). Accounts and sync use Supabase through its official client (ADR-012).
- **Documented architecture.** The repository must show a tier breakdown, C4 diagrams (Context, Container, Component), UML class and sequence diagrams, and Architecture Decision Records, and the code must follow them.
- **Code quality.** The code must use classes to bundle data with behavior, apply the object-oriented techniques that fit the problem, avoid duplication, and keep every stylesheet in its own imported file.
- **Semester timeline.** Work is planned in weekly sprints from August to November 2026.

1.3 Problem, Background, and Stakeholders

Collectors with hundreds of books regularly buy a second copy of something they already own, lose track of which edition they have, and forget which books on their shelves are still unread. Gifts are another blind spot: months later it is hard to remember who gave a book. General-purpose reading apps focus on reviews and social feeds; they do not model a physical collection, its editions, its duplicates, or how each copy was acquired.

The stakeholders are:

- **The bibliophile (primary user)**, who scans, shelves, and browses their own books.
- **The course instructor**, who evaluates the architecture, the code quality, and this document.
- **The peer review partner**, who must be able to run the app from the README and compare the diagrams to the code.
- **External services**: Open Library and Google Books (book data) and Supabase (accounts and the synced copy), whose public APIs the app depends on and whose terms it must respect.

1.4 Application Overview: Characteristics, Features, Differentiation, and Impact

The application is characterized by: (1) **cross-platform without native builds**, a single React Native (Expo) codebase that runs in the free Expo Go client on iPhone and Android, with Expo web as a conference fallback; (2) **exact-edition scanning**, where the ISBN barcode is read with the camera and the edition's own publisher, date, page count, binding, and cover are looked up, with the owner choosing the cover that matches their copy or photographing it; and (3) **collection rules in the domain model**, so that the app, not the user, guarantees that the catalog stays consistent.

The core features are:

1. **ISBN scan** with the camera, or a typed ISBN when no camera is available.
2. **Owned and Wanted shelves**. "To be read" (TBR) is a view of Owned books split into Unread and Read, not a third shelf.
3. **Owned catalog** grouped A–Z, by author, or by series, with an A–Z letter rail for fast scrolling.
4. **Domain rules**: scanning an ISBN that is already cataloged adds a copy instead of a second entry; a book cannot be Owned unless it was purchased or gifted; a gift requires the giver's name; reading status is independent of the shelf.
5. **Book detail** with edition facts, binding, read/unread toggle, shelf moves, a cover photo option, and deletion with confirmation.
6. **Accounts and sync**: email sign-in, and the library kept in sync on every phone signed in to the same account.

Unlike reading-tracker apps, MyHomeLib records the physical copy: which edition and binding it is, how many copies the owner has, and whether each one was bought or received as a gift. It runs on the user's device and works offline, so the library stays private; account sync (feature 4) keeps a copy in the owner's own account, which row level security keeps from every other account; it is not a shared catalog. The expected impact is fewer duplicate purchases, a reliable to-be-read list, and a record of gifts.

1.5 Scope

This document covers the final version of MyHomeLib. The system consists of four tiers (Presentation, Service, Domain, Data) and seven major components: the Presentation view models, the Service use cases, the Domain model, the book repositories, the book-source adapters, the account adapters, and the device adapters (camera and haptics). Each component is described in Section 3. The scope for the final is frozen to the four core features in Section 1.1: all four are built and tested. Out of scope: sharing between users, price tracking, bookstore integrations, and anything else not listed as a core feature. Three scope changes are acknowledged against the approved proposal: the bookstore price check was removed (ADR-004); Google Books was added as an optional second metadata source so that

scans match the exact edition (ADR-006), through its official API, with the app still working without it; and account sign-in with library sync was added as the fourth core feature when the scope was frozen at the midterm check-in. It was planned as Google sign-in and built with email accounts through Supabase, because Google sign-in needs a custom native build that is not free on iPhone (ADR-012); password reset is left out for now, because on Supabase's free plan the reset email cannot carry a code the app could use.

1.6 Definitions, Acronyms and Abbreviations

Term	**Meaning**
ADR	Architecture Decision Record: a one-page record of a design decision, its context, and its consequences
API	Application Programming Interface
C4	Context, Container, Component, Code: a model for diagramming software architecture at four zoom levels
Expo Go	Free client app that runs an Expo (React Native) project on a phone without a native build
ISBN	International Standard Book Number; ISBN-13 barcodes are EAN-13
MVVM	Model–View–ViewModel: a presentation pattern in which an observable view model holds screen state and logic
OOP	Object-Oriented Programming
Port	An interface owned by the Service tier that the Data tier implements
Realtime	Supabase's live change notices, which tell a phone that its account's rows changed
RLS	Row level security: database rules that let a signed-in account read and write only its own rows
SDD	Software Design Document
Supabase	Hosted service (Postgres database, sign-in, Realtime) that holds the accounts and each account's synced copy of the library
TBR	"To be read": Owned books, split into Unread and Read
Tombstone	A deleted book's row kept with its deletion time, so the deletion syncs to other phones
UML	Unified Modeling Language
WCAG	Web Content Accessibility Guidelines

1.7 Acknowledgements

The requirements, domain rules, feature decisions, and design direction were set by Grace Hechavarria, who reviewed and approved every change. Anthropic's Claude (Opus 5.5), used through the Claude Code tool (Anthropic, 2026), assisted with implementation, refactoring toward the class-based architecture, automated tests, rendering the Mermaid diagrams to images, capturing the application screenshots, and drafting and assembling this document. Design-review skill packages (Anthropic frontend-design, TasteSkill, UI/UX Pro Max, Vercel React Native guidelines, AccessLint, and the Bencium UX designer guide) were used as checklists during the interface redesign. The document was typeset with LaTeX (MiKTeX). The structure of this SDD follows a template from a previous FIU software engineering course. Thanks to the peer review partner, Yousef Habouh, for the structured reviews.

1.8 References

- Anthropic. (2026). *Claude* (Opus 5.5 model) [Large language model], accessed through *Claude Code* [Computer software]. Retrieved from https://www.anthropic.com/claude-code
- Expo. (n.d.). *Expo documentation*. Retrieved from https://docs.expo.dev/
- Internet Archive. (n.d.). *Open Library APIs*. Retrieved from https://openlibrary.org/developers/api
- Google. (n.d.). *Books API*. Retrieved from https://developers.google.com/books
- Brown, S. (n.d.). *The C4 model for visualising software architecture*. Retrieved from https://c4model.com/
- Supabase. (n.d.). *Supabase documentation*. Retrieved from https://supabase.com/docs

2 System Overview

Figure 1 shows MyHomeLib in its environment. A single type of user, the bibliophile, catalogs books with the app. The app depends on three external systems: Open Library, which needs no key, and Google Books, which needs a free API key, for book metadata; and Supabase, which holds the accounts and each account's synced copy of the library. Everything else runs on the user's device, and the device's copy keeps working offline.

Figure 2 shows the architecture chosen for MyHomeLib: a **four-tier modular monolith** in one TypeScript codebase. The tiers are folders with a strict dependency direction, enforced by automated tests:

- **Presentation** (`presentation/`): React Native screens rendered by Expo, the `LibraryAppModel` view model that holds all screen state and user flows, and `AccountModel` for the Account tab. Components only render and forward taps.

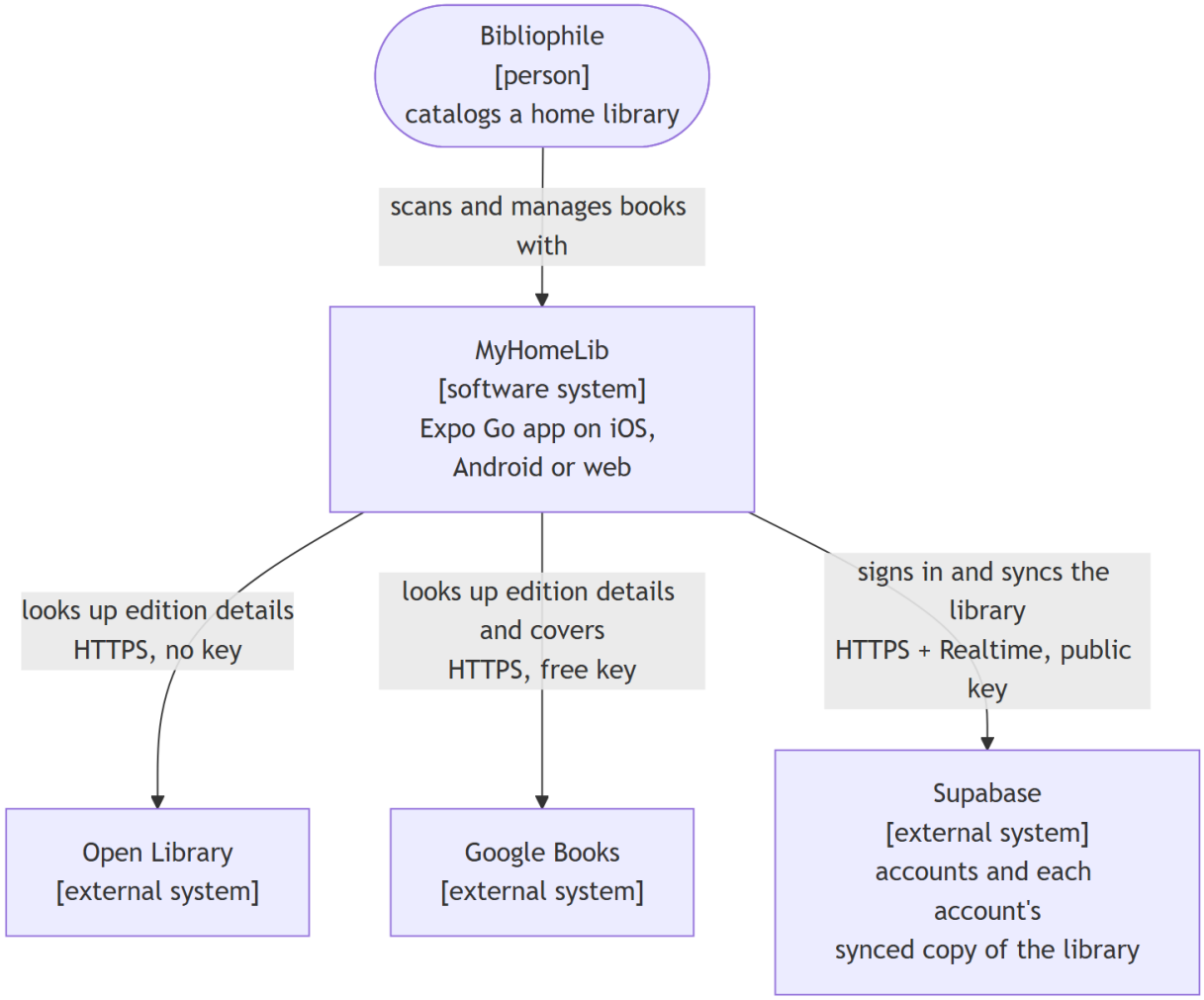


Figure 1: C4 Level 1 – System Context

- **Service** (`service/`): one class per use case (scan a book, move a book, set reading status, sign in, sync the library, and so on) plus the *ports*, the interfaces that the Data tier implements.
- **Domain** (`domain/`): the business objects and rules: `Book`, `PersonalLibrary`, and the acquisition records `Purchase` and `Gift`. Domain imports nothing outside itself.
- **Data** (`data/`): the adapters to the outside world: SQLite on a phone, browser storage on the web, the Open Library and Google Books lookups (merged and cached), and the Supabase sign-in and backup.

Calls always flow Presentation → Service → Domain and ports, never backwards. The Data tier depends on the Service ports, not the other way around, so the business rules can be tested with in-memory fakes and the storage technology can change without touching them. A composition root (`bootstrap.ts`) is the only place that chooses concrete adapters and wires them into the use cases.

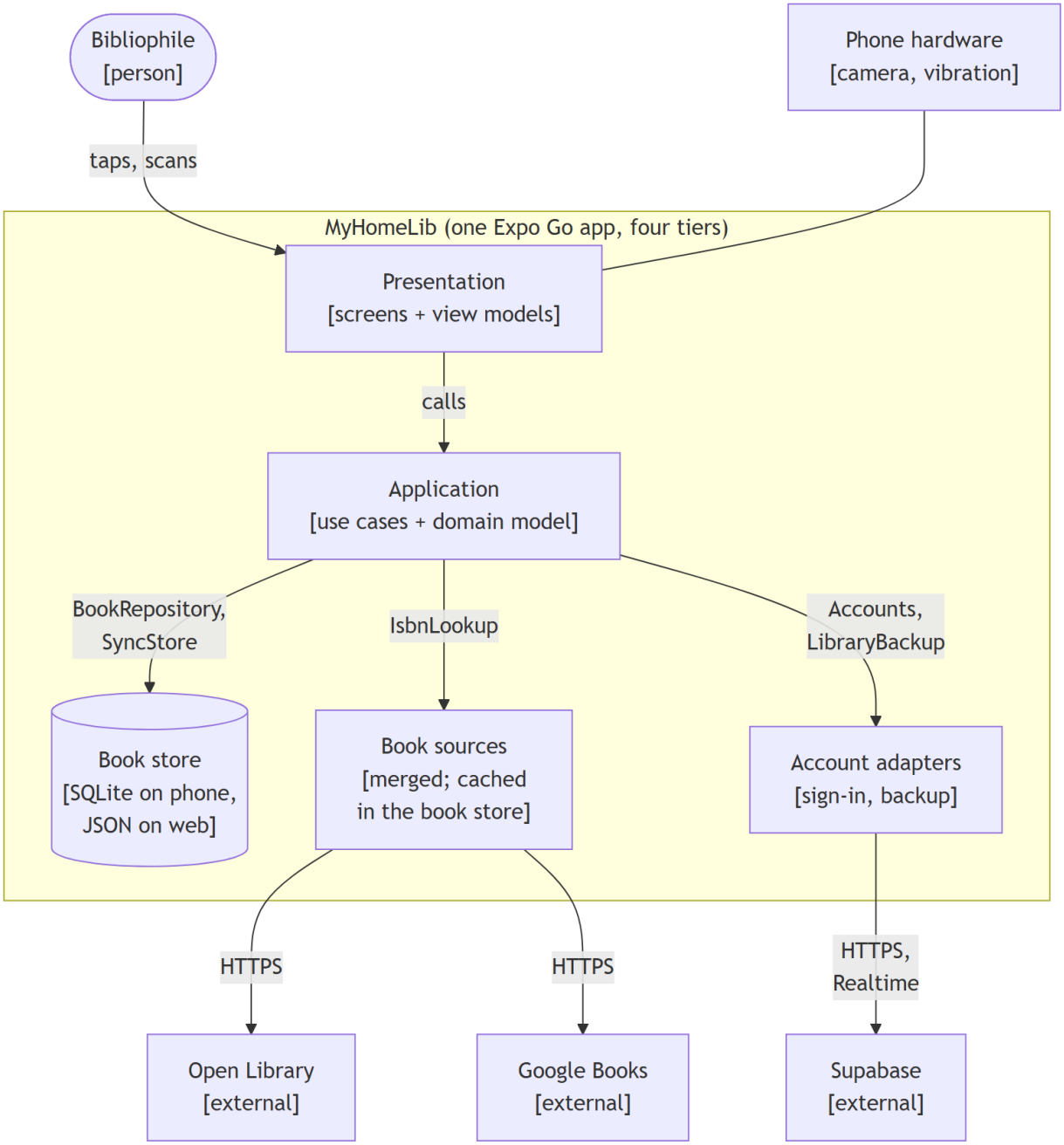


Figure 2: C4 Level 2 – Container (System Architecture)

Because MyHomeLib runs in Expo Go, the same code reaches iOS, Android, and the web. Feature 4, account sign-in and library sync, fits this architecture: three new ports (`SyncStore`, `LibraryBackup`, `Accounts`) with Supabase adapters, and every book already stored per owner, so the library use cases and the domain rules did not change (ADR-012).

3 System Components

3.1 Decomposition Description

Figure 3 shows a top-down decomposition of the codebase. Each tier is a folder, and each folder holds classes with one responsibility each:

The **Presentation view model** (`presentation/model/`) is the controller of the C4 component view. `LibraryAppModel` extends a reusable `Observable` base class and holds every screen's state: the selected tab, the open popup, the scan draft, and the flash message. `LibraryController` turns screen intents into use case calls and returns a `LibraryView` snapshot. Helper classes keep logic out of the views: `ScanDraft` (a scan in progress), `Catalog` and its grouping strategies, `AlphabetRail` (letter-rail math), `Labels` (all display wording), and `PromptActions` (popup buttons). `AccountModel`, also an `Observable`, holds the Account tab (the form, the signed-in account, and how the last sync went); `LibraryAppModel` follows it, showing the signed-in account's books and an empty, locked library when nobody is signed in.

The **Presentation views** (`presentation/components/` and `presentation/styles/`) are React components that only render. Every component imports its stylesheet from `styles/`, and every color, size, and font comes from the theme tokens. Shared building blocks (`Tappable`, `TextField`, `BookCover`, `CoverBackdrop`, `Logo`, `Reveal`) replace repeated code.

The **Device families** (`presentation/device/`) wrap the device: `PhoneDevices` creates the real camera (`ExpoCoverCamera`) and haptics (`HapticFeedback`); `BrowserDevices` creates a camera that points to the phone app and silent feedback. Both implement the `DeviceFamily` interface (Abstract Factory, ADR-011).

The **Service use cases** (`service/`) orchestrate one user goal each. Eight library use case classes are built once by the `UseCases` class from two ports; the sign-up, sign-in and sign-out use cases and `SyncLibraryUseCase` are built by `AccountUseCases` from three more.

The **Domain model** (`domain/`) enforces the collection rules. Its state is private and changes only through methods that check the rules.

The **Book repositories** (`data/`) persist the library: `SqliteBookRepository` on a phone, `JsonBookRepository` on the web, and `MemoryBookRepository` for tests, all translating rows through `BookRowMapper`. Each also serves sync: every save is stamped with its time, and a delete leaves a tombstone.

The **Book-source adapters** (`data/`) look up an ISBN: `OpenLibraryIsbnLookup` and `GoogleBooksIsbnLookup`, combined by `MergedIsbnLookup` and cached by `CachingIsbnLookup`. All HTTP goes through one `Http` class.

The **Account adapters** (`data/`) reach Supabase: `SupabaseConnection` creates the one client (the session kept in AsyncStorage), `SupabaseAccounts` signs in and ex-

plains every failure in plain words, and `SupabaseLibraryBackup` reads and writes the `library_books` table in pages and watches it through Realtime. `UnconfiguredAccounts` and `UnconfiguredLibraryBackup` are Null Objects for a build with no Supabase settings.

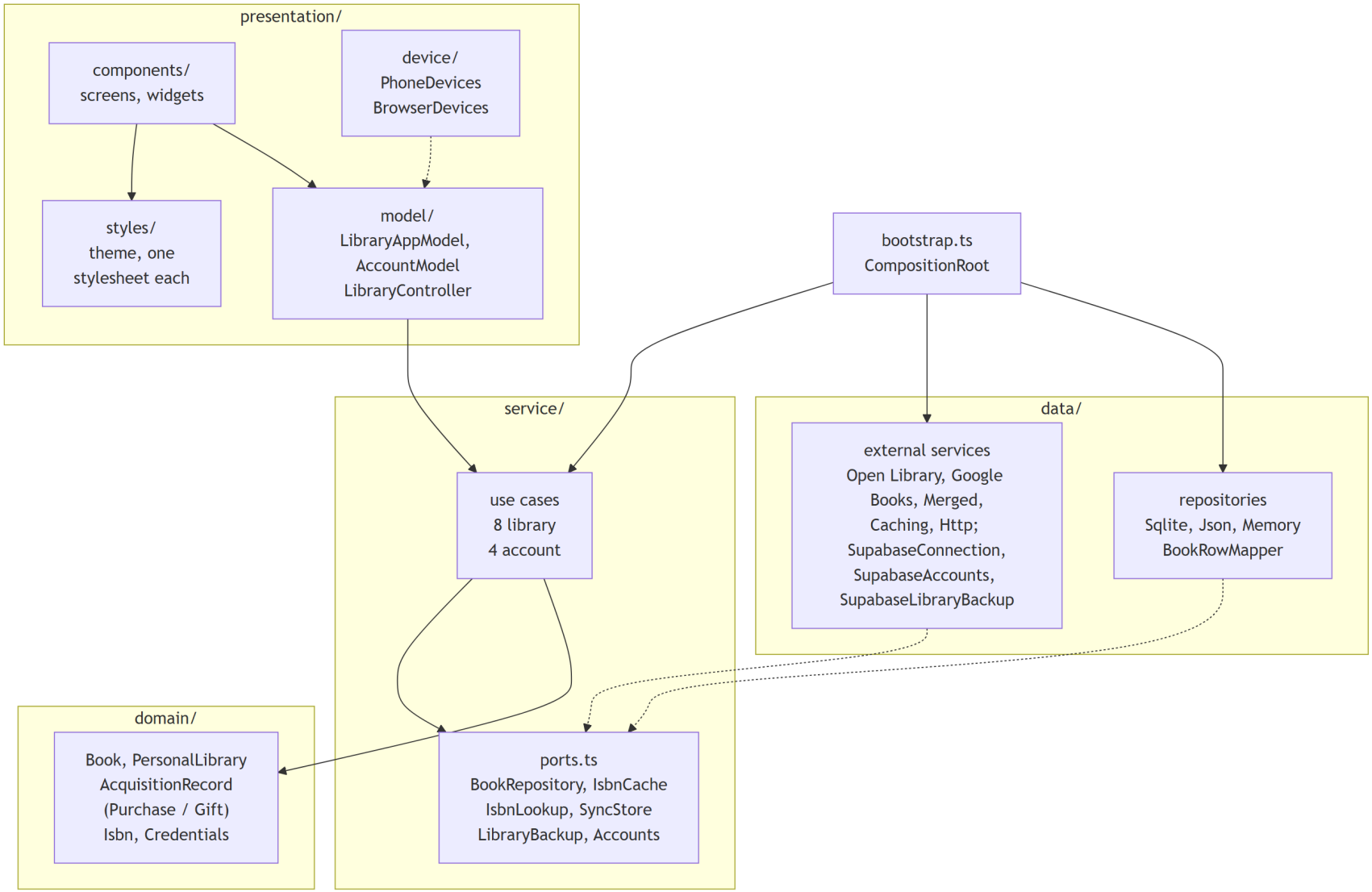


Figure 3: System Decomposition Diagram (folders and their main classes; a dotted arrow means "implements"; the text below lists every class)

3.2 Dependency Description

Figure 4 shows how each module depends on the others. `App.tsx` creates the `LibraryAppModel`, which uses `LibraryController` to reach the `UseCases`. The use cases depend on the Domain classes and on two *interfaces*, `BookRepository` and `IsbnLookup`. The concrete repositories and lookups depend on those interfaces (they implement them), which inverts the usual direction: Service never imports Data (the Dependency Inversion Principle, ADR-009). `AccountModel` reaches `AccountUseCases` the same way; their ports (`Accounts`, `LibraryBackup`, `SyncStore`) are implemented by the Supabase adapters and the device stores.

`CompositionRoot` in `bootstrap.ts` is the only module that knows every concrete class. At start-up it opens SQLite (or browser storage when SQLite is unavailable), wraps the merged book sources in the cache, builds `AccountUseCases` with the Supabase adapters (the project named in `.env`, or the shared one built into the app), and hands both to the view models. The device services follow the same idea one tier up: the view model depends on the `DeviceFamily` interface, `App.tsx` passes `PhoneDevices` or `BrowserDevices`, and tests pass a fake family.

These rules are not only documented; `tests/architecture.test.ts` reads every source file and fails if Domain imports anything outside Domain, if Service imports Data or calls `fetch`, if Data imports Presentation or React, if Presentation reaches a repository, if a component defines its own stylesheet, or if any file other than `App.tsx` reaches the composition root.

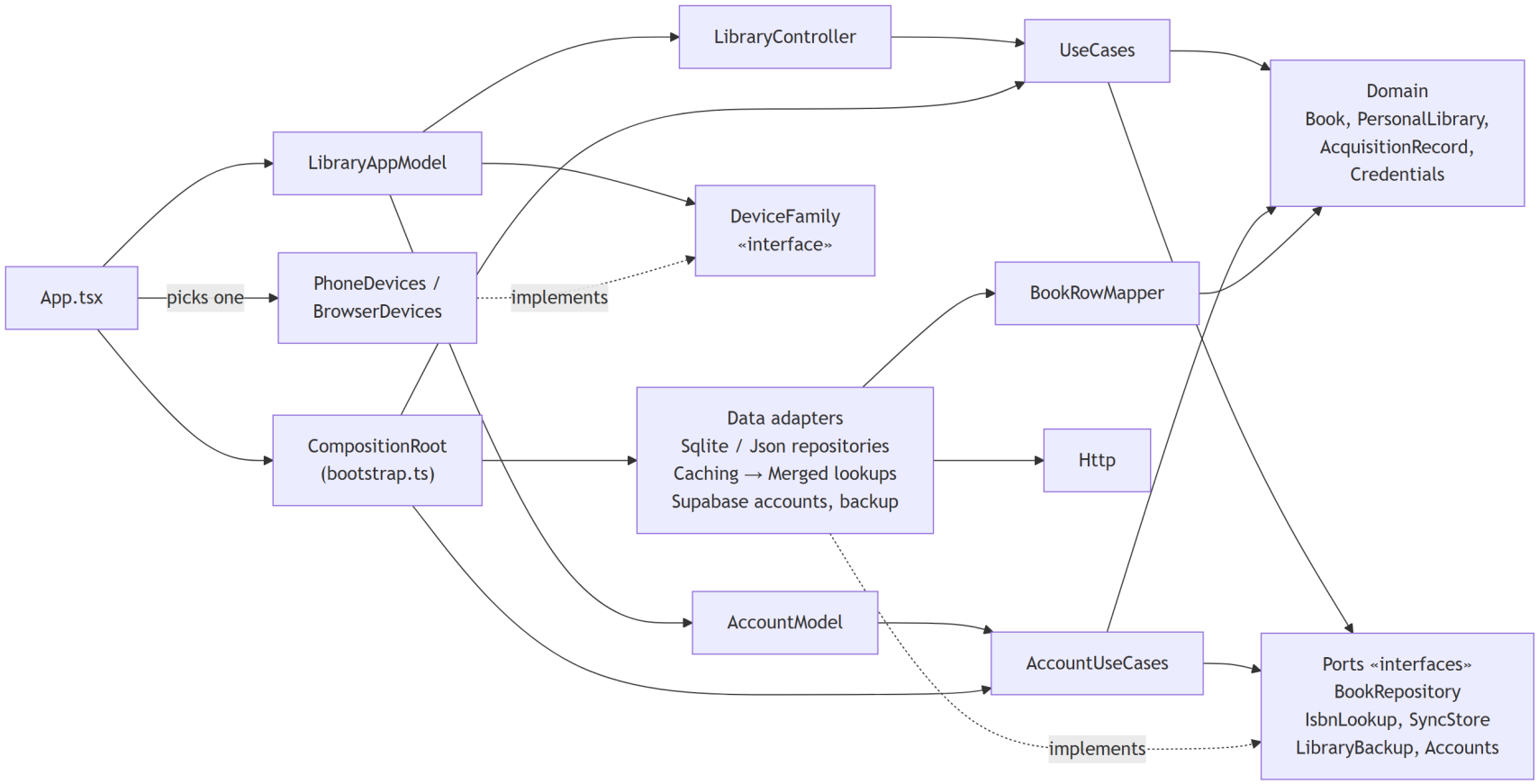


Figure 4: System Dependency Diagram (the Data adapters are grouped; Figures 8, 9 and 10 show each class)

3.3 Interface Description

3.3.1 Presentation View Model

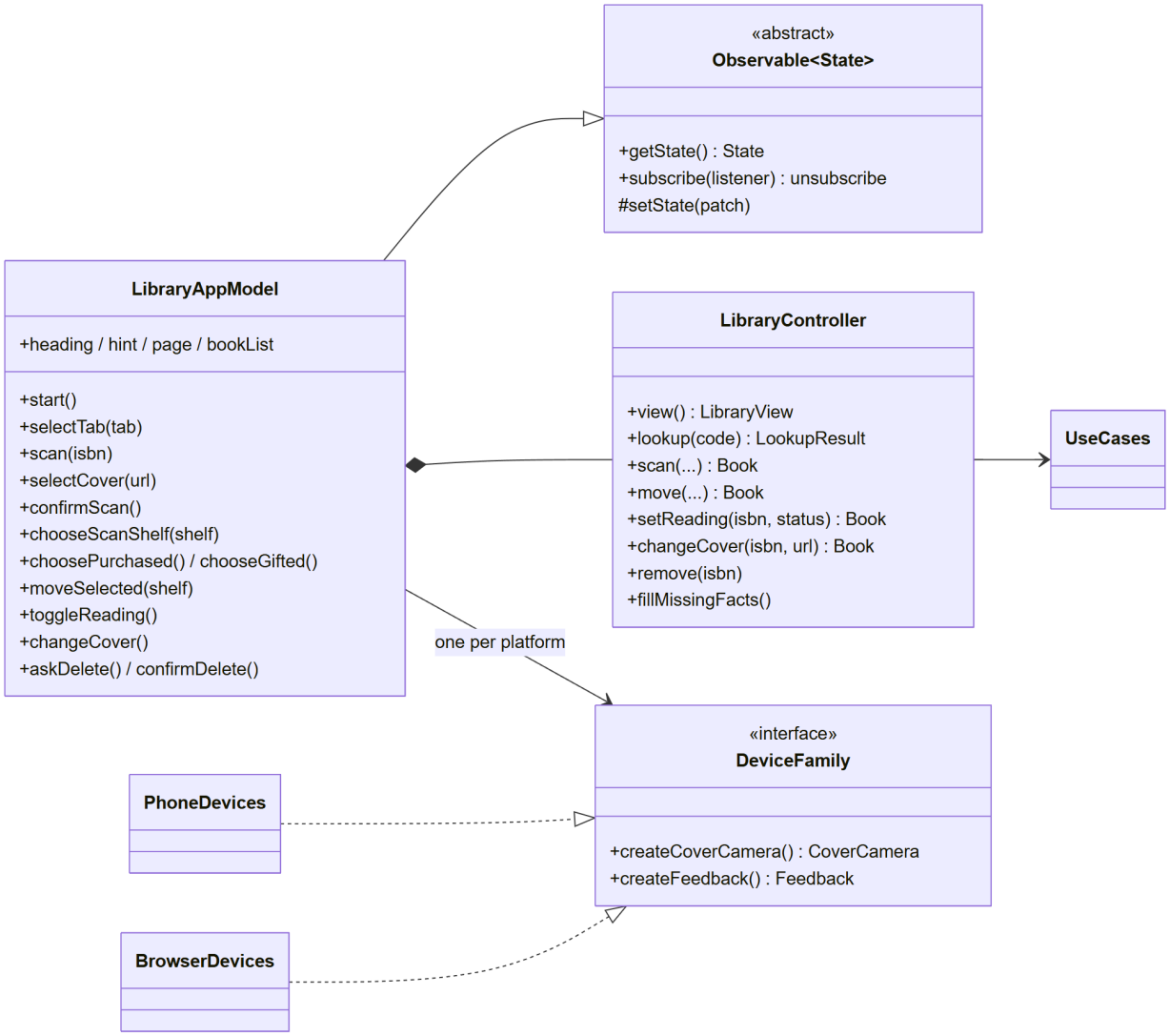


Figure 5: Presentation View Model Interface

Figure 5 shows the interface between the screens and the rest of the app. Screens read `LibraryAppModel`'s state and derived values (`heading`, `hint`, `page`, `bookList`) and call its intent methods (`scan`, `confirmScan`, `chooseGifted`, `moveSelected`, ...). React subscribes through one hook, `useObservable`, so every state change re-renders the screens. The model calls the use cases only through `LibraryController`, and gets its camera and feedback only from the `DeviceFamily` it is given.

3.3.2 Service Use Cases

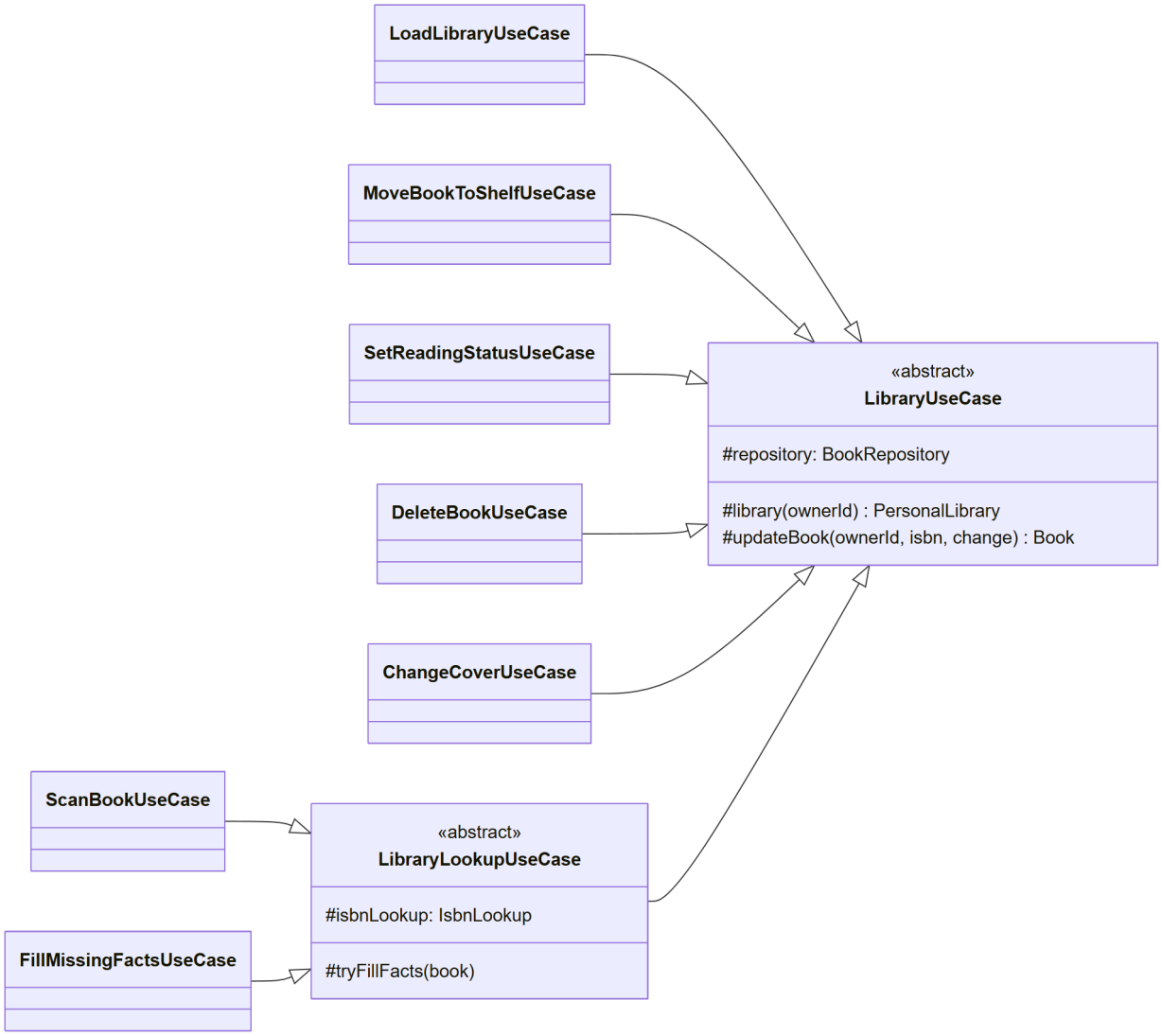


Figure 6: Service Use Case Interface

Figure 6 shows the use case classes. Every library use case extends the abstract `LibraryUseCase`, which implements the shared load → change → save steps once (`updateBook`, a Template Method). Use cases that also need the book sources extend `LibraryLookupUseCase`. `LookupIsbnUseCase` only reads, so it stands alone. Each use case exposes a single `execute(...)` method.

3.3.3 Domain Model

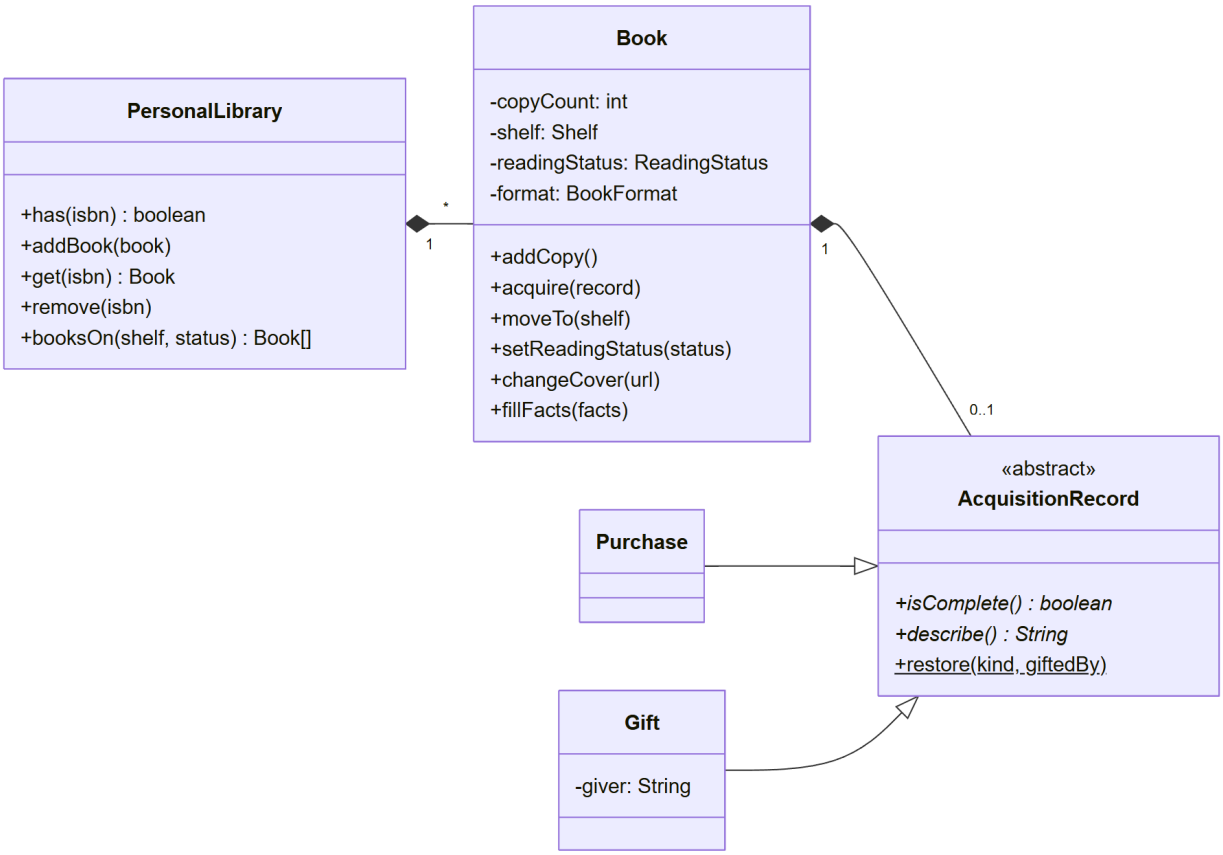


Figure 7: Domain Model Interface

Figure 7 shows the domain interface. `PersonalLibrary` composes the `Book` objects of one owner. `Book` keeps its fields private and offers intention-revealing methods (`addCopy`, `acquire`, `moveTo`, ...) that throw a domain error when a rule is broken. `AcquisitionRecord` is abstract; `Purchase` and `Gift` implement `isComplete()` and `describe()` polymorphically, so `Book` never branches on the acquisition type.

3.3.4 Book Repositories

Figure 8 shows the persistence interface. `BookRepository` (the library) and `IsbnCache` (looked-up metadata) are separate ports, so the use cases depend only on the library methods they call and a store with no reason to cache lookups, such as the account backup, need not implement the cache (Interface Segregation). `SyncStore` is the sync side: every stored version of a book, tombstones included, and a way to apply versions that keeps their timestamps and never replaces a newer copy. The three device stores implement all three, so one store holds the library, the cached lookups, and what sync needs. `JsonBookRepository` reuses `MemoryBookRepository` by composition and only adds loading and saving to browser storage. `BookRowMapper` is the single translation between a stored row and a `Book` (Data Mapper pattern), including the defaults for rows saved by older versions.

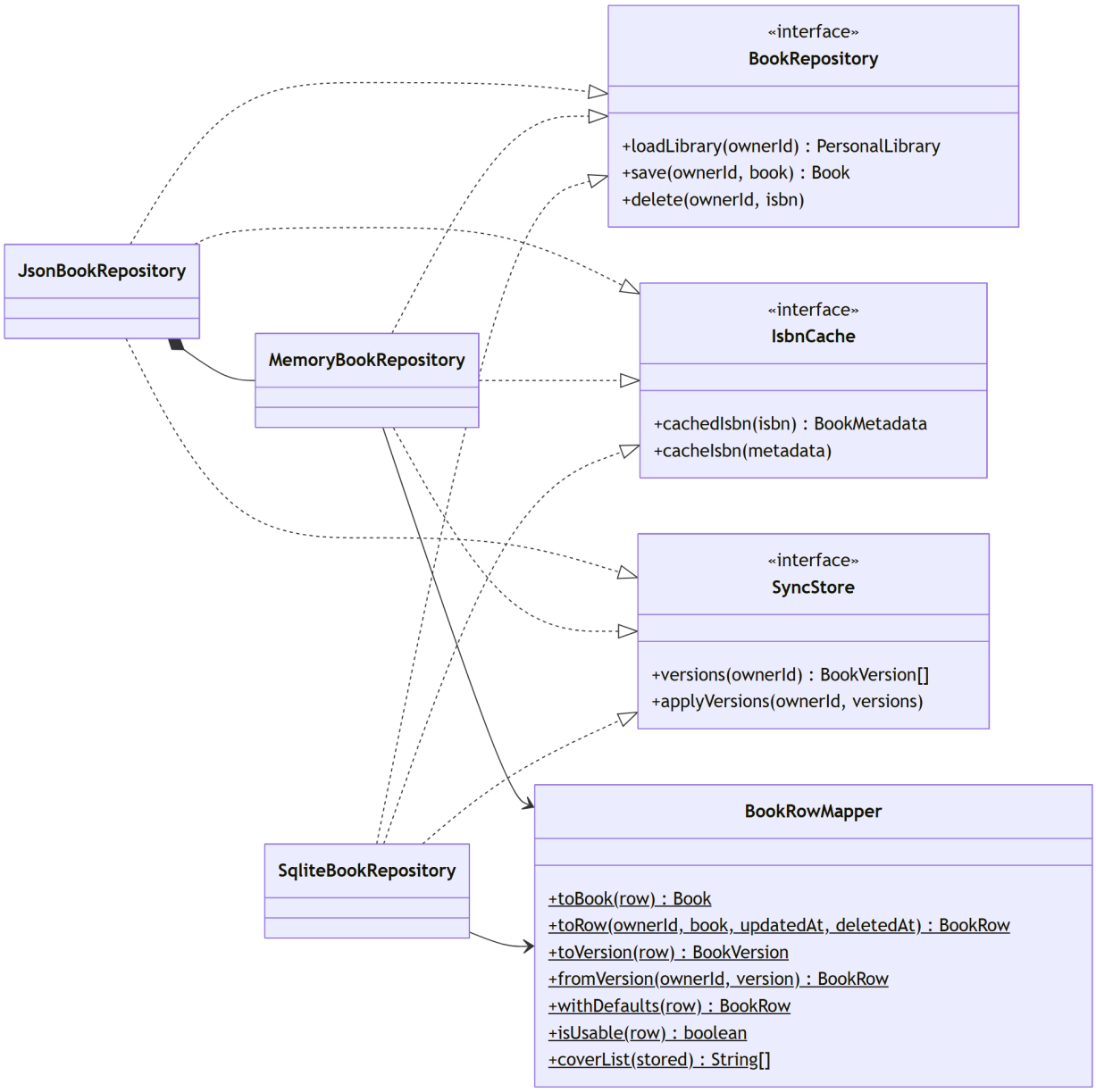


Figure 8: Book Repository Interface

3.3.5 Book-Source Adapters

Figure 9 shows the lookup chain. Every element implements the same `IsbnLookup` interface, so they stack as decorators: `CachingIsbnLookup` answers from the cache or asks `MergedIsbnLookup`, which asks Google Books and Open Library together, merges the answers field by field (edition data first), and keeps every distinct cover as a candidate. Both sources reach the network only through the `Http` class.

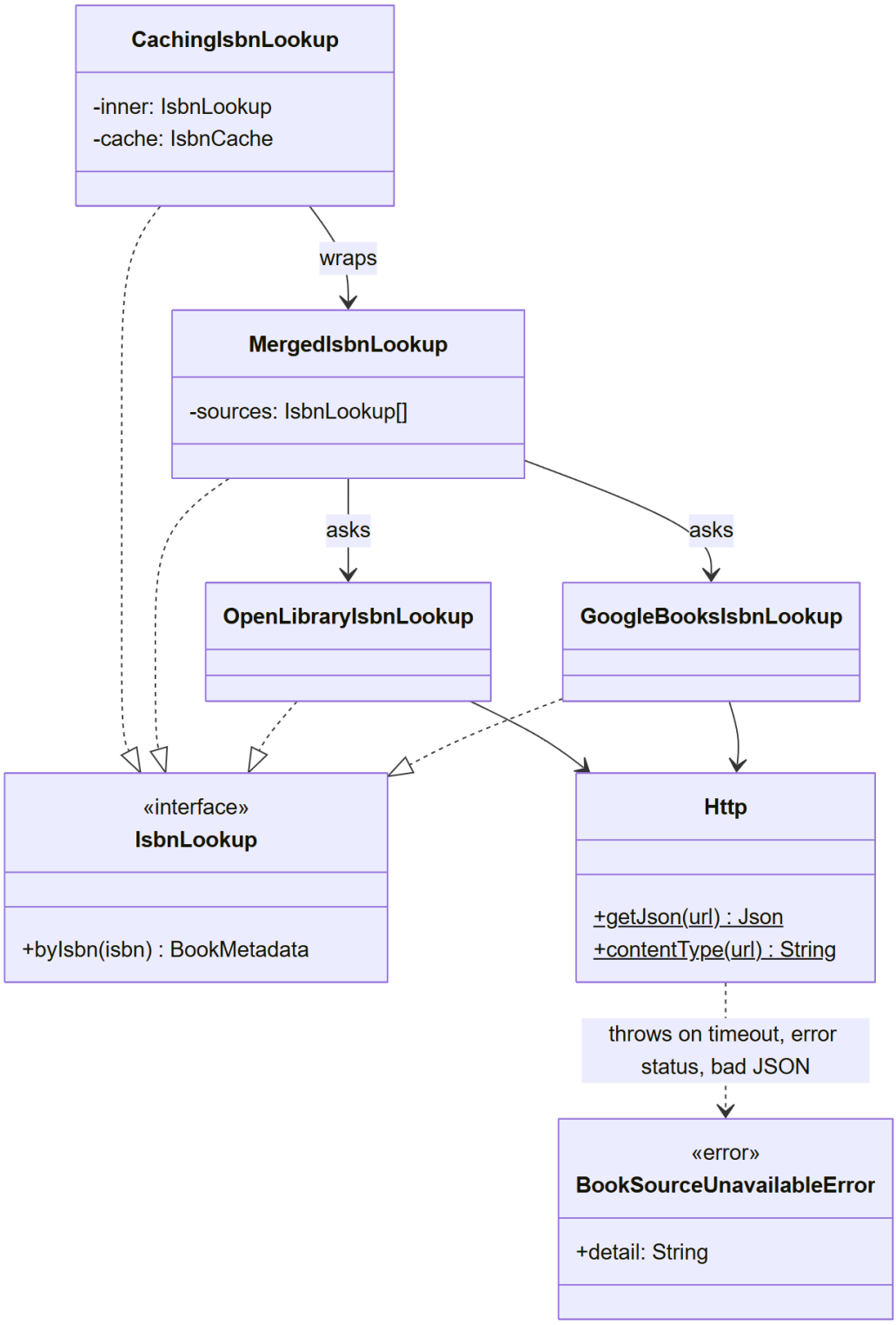


Figure 9: Book-Source Adapter Interface

3.3.6 Accounts and Sync

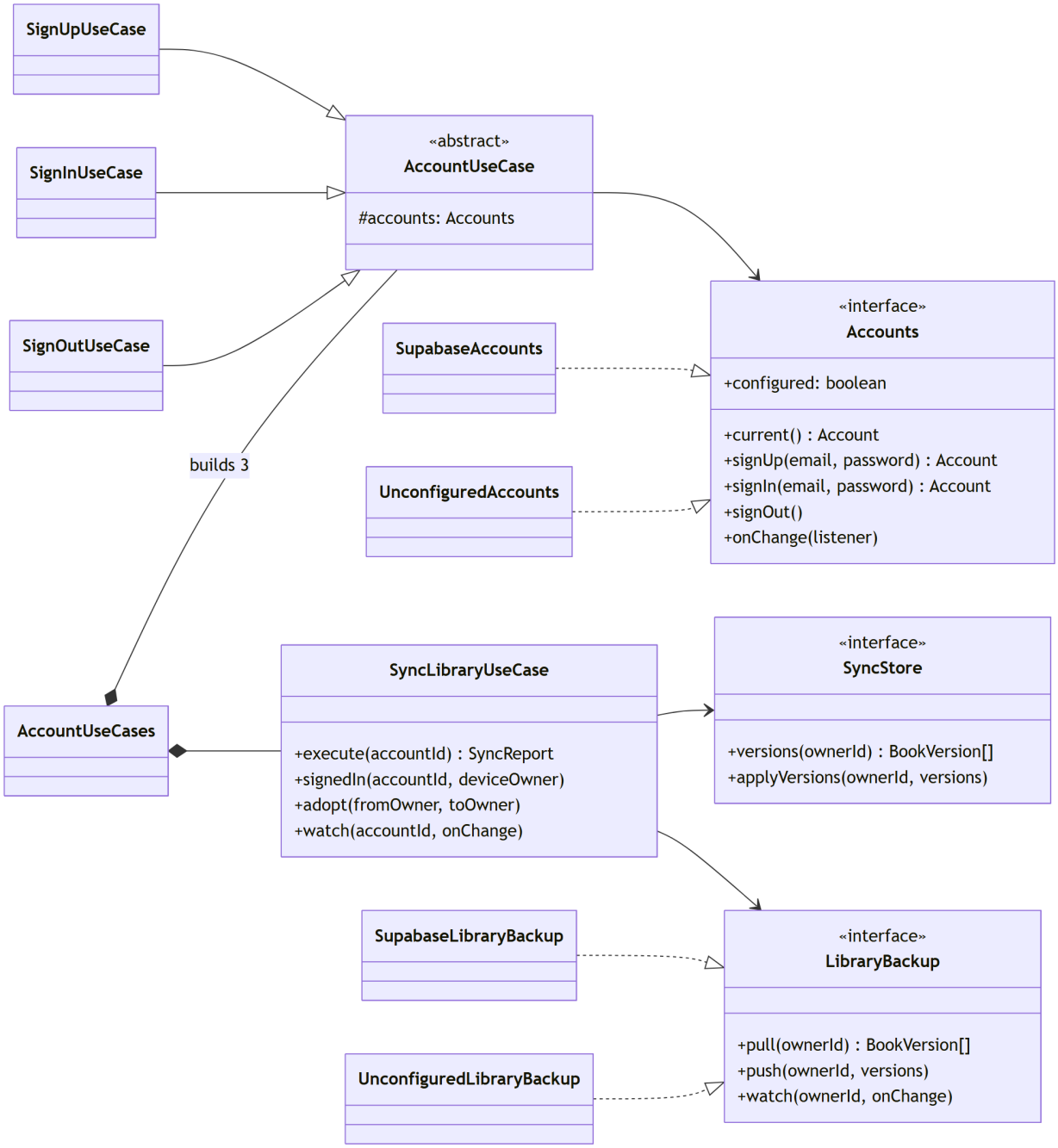


Figure 10: Accounts and Sync Interface (ADR-012)

Figure 10 shows feature 4. The account use cases extend `AccountUseCase`, which holds the `Accounts` port; each checks the email and password in the domain (`Credentials`) before any network call. `SyncLibraryUseCase` compares the phone's versions (`SyncStore`) with the account's (`LibraryBackup`) book by book: the newer `updatedAt` wins in both directions, and on the first sign-in the books kept on the phone move into the account, a book in both

adding its copies. Supabase implements both account ports; the Null Objects stand in when the app has no Supabase settings.

3.3.7 Device Adapters

The camera and the vibration motor come from one device family, so the view model never touches device APIs directly and a phone service is never mixed with a browser one (Abstract Factory, ADR-011). `App.tsx` picks the family once. On a phone, `ExpoCoverCamera` opens the camera, lets the owner crop to the cover, and shrinks the photo to a small JPEG stored with the book, and `HapticFeedback` gives a light tick when a tab or switch changes, a soft tap for a message, and a buzz for an error. In a browser, the camera explains that it needs the phone app and the feedback is silent. The view model triggers the feedback in the same actions that change the screen. The ISBN barcode scanner is `IsbnCamera`, a view that reports each new code to the view model.

3.4 Module Interfaces

The following TypeScript interfaces are the contracts between tiers. The Service tier owns them (`service/ports.ts`); the Data tier implements them.

```
/** What the book sources know about one ISBN (one edition). */
export type BookMetadata = BookFacts & {
  isbn: string; title: string; author: string; publisher: string;
  coverUrl: string; // best cover first
  coverCandidates: string[]; // every distinct cover the sources offer
};

/** The one error a source throws when it can't answer (offline, timeout, 429,
    bad JSON). */
export class BookSourceUnavailableError extends Error {
  constructor(readonly detail: string);
}

export interface IsbnLookup {
  byIsbn(isbn: string): Promise<BookMetadata>;
}

export interface IsbnCache {
  cachedIsbn(isbn: string): Promise<BookMetadata | null>;
  cacheIsbn(metadata: BookMetadata): Promise<void>;
}

/** Kept separate from IsbnCache: a library store need not cache lookups. */
export interface BookRepository {
  loadLibrary(ownerId: string): Promise<PersonalLibrary>;
  save(ownerId: string, book: Book): Promise<Book>;
  delete(ownerId: string, isbn: string): Promise<void>;
}
```

```ts
/** Account sync (ADR-012): versions are compared book by book. */
export type BookVersion = { book: Book; updatedAt: number; deletedAt: number |
    null };

export interface SyncStore { // the device store's sync side
  versions(ownerId: string): Promise<BookVersion[]>;
  applyVersions(ownerId: string, versions: BookVersion[]): Promise<void>;
}

export interface LibraryBackup { // the account's copy (Supabase)
  pull(ownerId: string): Promise<BookVersion[]>;
  push(ownerId: string, versions: BookVersion[]): Promise<void>;
  watch(ownerId: string, onChange: () => void): () => void;
}

export interface Accounts {
  readonly configured: boolean;
  current(): Promise<Account | null>;
  signUp(email: string, password: string): Promise<Account | null>;
  signIn(email: string, password: string): Promise<Account>;
  signOut(): Promise<void>;
  onChange(listener: (account: Account | null) => void): () => void;
}
```

The Presentation tier owns two more contracts: the device camera, and the reusable observable base class of the view model.

```ts
export interface CoverCamera {
  readonly available: boolean; // false in the browser build
  // A small image of the cover (data URI), or null if the owner backed out.
  takePhoto(): Promise<string | null>;
}

export interface Feedback {
  selection(): void; // a tab or switch changed
  message(failed: boolean): void; // a banner appeared
}

/** Abstract factory: the device services that belong together. */
export interface DeviceFamily {
  createCoverCamera(): CoverCamera;
  createFeedback(): Feedback;
}

export abstract class Observable<State> {
  getState(): State;
  subscribe(listener: () => void): () => void; // returns unsubscribe
```

```
  protected setState(patch: Partial<State>): void;
}
```

3.5 User Interfaces (GUI)

The interface uses one design system: a dark "enchanted forest" background, a single mint accent reserved for selection and primary actions, the Outfit typeface in four weights, and a floating tab dock with TBR, Owned, Wanted, Scan, and Account. All screenshots below were captured from the running application at phone size. The Account tab (feature 4) shows the sign-in or create-account form, or, once signed in, the account with *Sync now* and *Sign out*; while nobody is signed in, TBR, Owned, Wanted and Scan show one "Sign in to see your library" card that opens it.

3.5.1 To Be Read

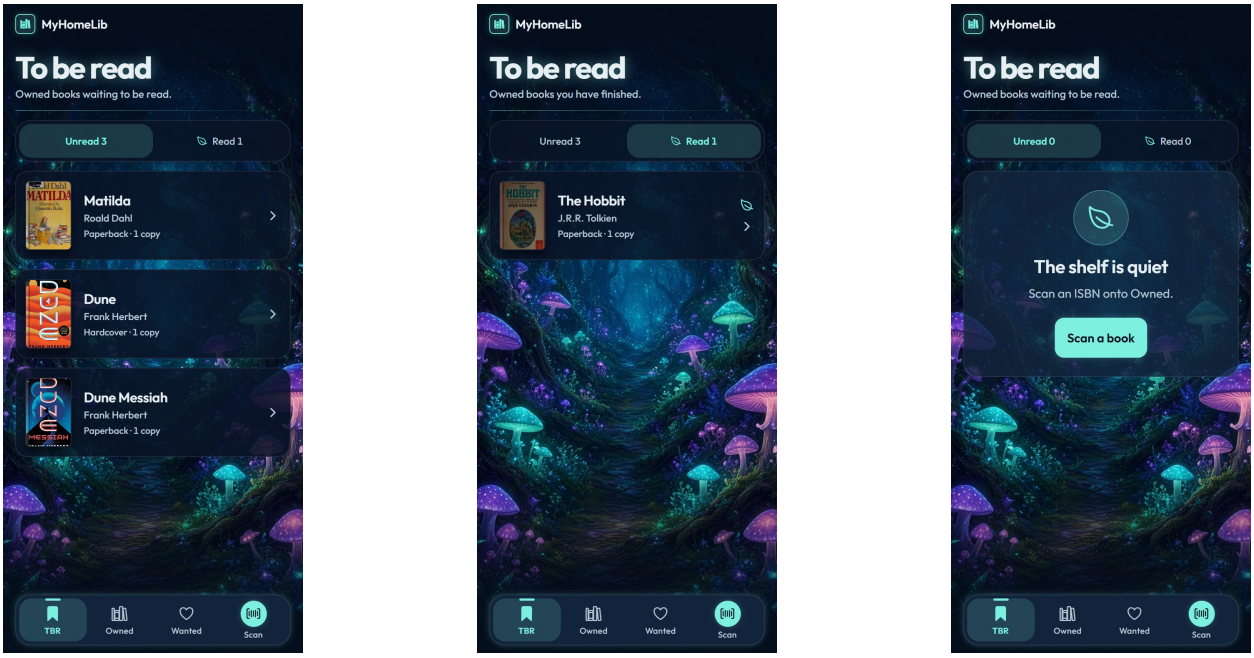


Figure 11: To be read: Unread (left), Read (center), and the empty state (right)

Figure 11 shows the TBR screen, the app's home. The switch at the top splits Owned books into Unread and Read, with a count on each side. Each card shows the cover, title, author, binding, and copy count; a leaf marks books already read. An empty shelf explains itself and offers a Scan button.

3.5.2 Owned Catalog

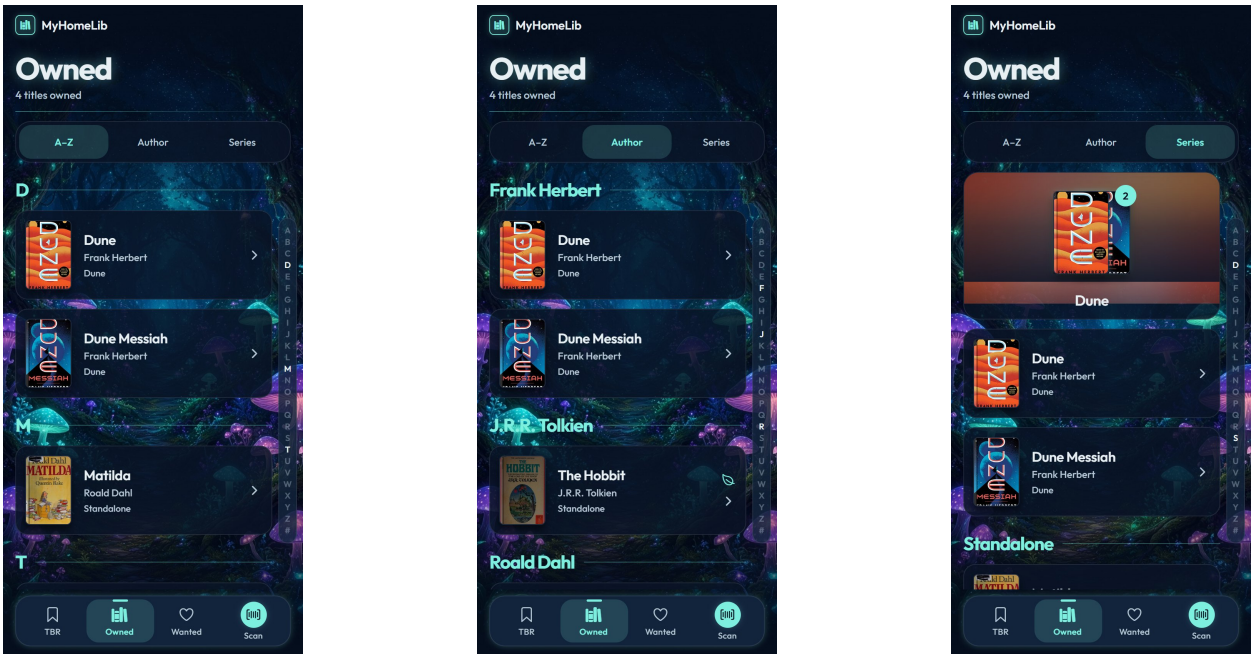


Figure 12: Owned catalog grouped A–Z (left), by author (center), and by series (right)

Figure 12 shows the full Owned catalog. The switch chooses the grouping; each section starts with an index heading. In the series grouping, a series appears as a fan of its covers with a count badge, and tapping it expands its books. The A–Z rail on the right jumps to a section when tapped or dragged.

3.5.3 Wanted

Figure 13 shows the wish list. A Wanted book has the same detail screen as an Owned one, with a “To Owned” action instead of “To Wanted”.

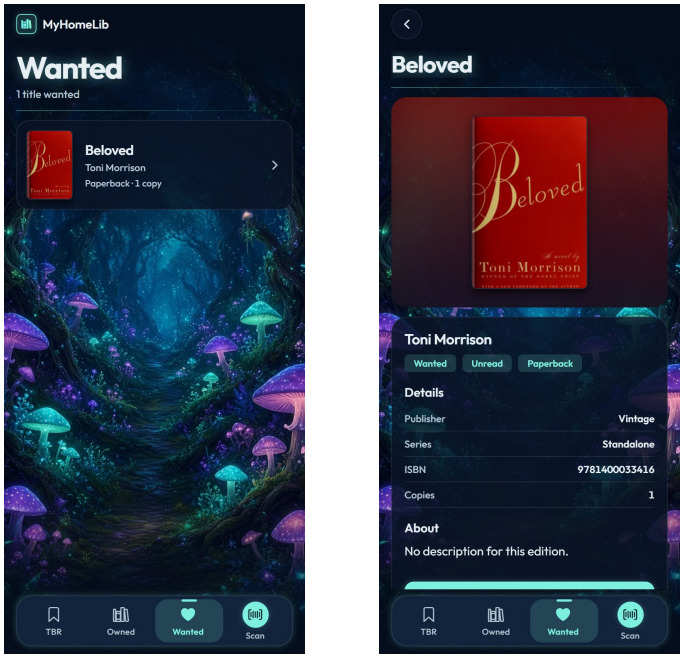


Figure 13: Wanted list (left) and a Wanted book's detail (right)

3.5.4 Scan

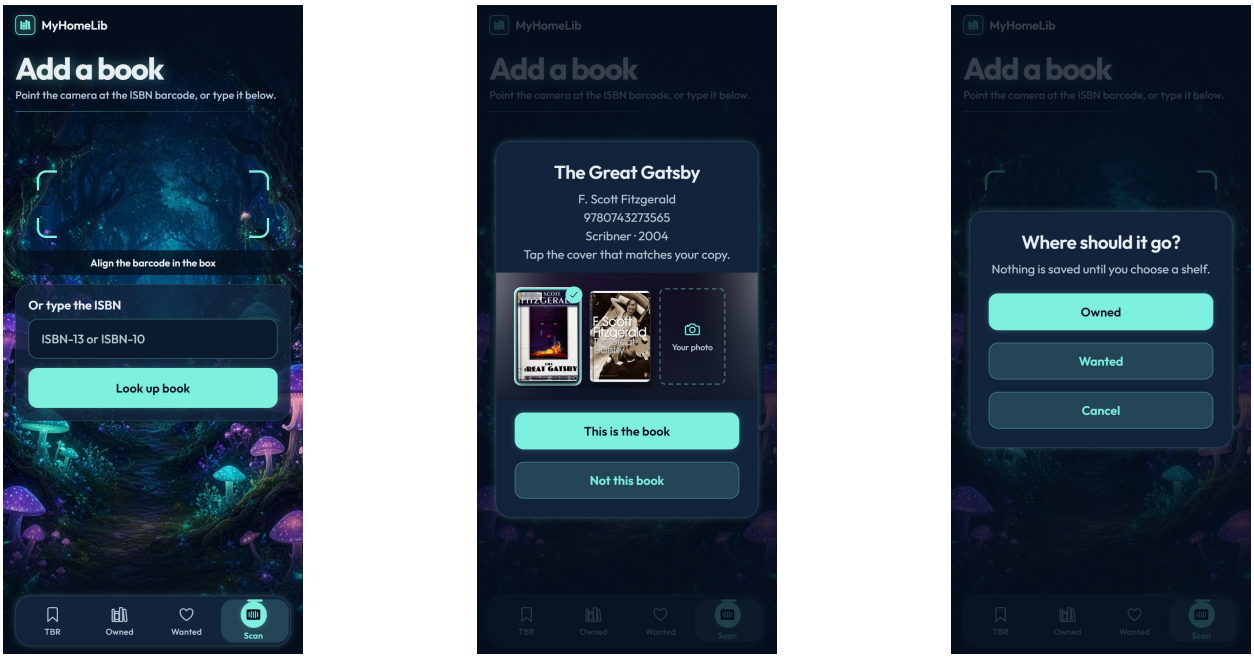


Figure 14: Scan screen (left), the confirm popup with the cover picker (center), and the shelf choice (right)

Figure 14 shows the scan flow. The camera reads the barcode inside the corner brackets; the ISBN can also be typed. The confirm popup shows the edition's details and every cover the sources offer, on a soft halo of the selected cover, plus a "Your photo" tile. Nothing is

saved until the owner has answered every question.

3.5.5 Book Detail

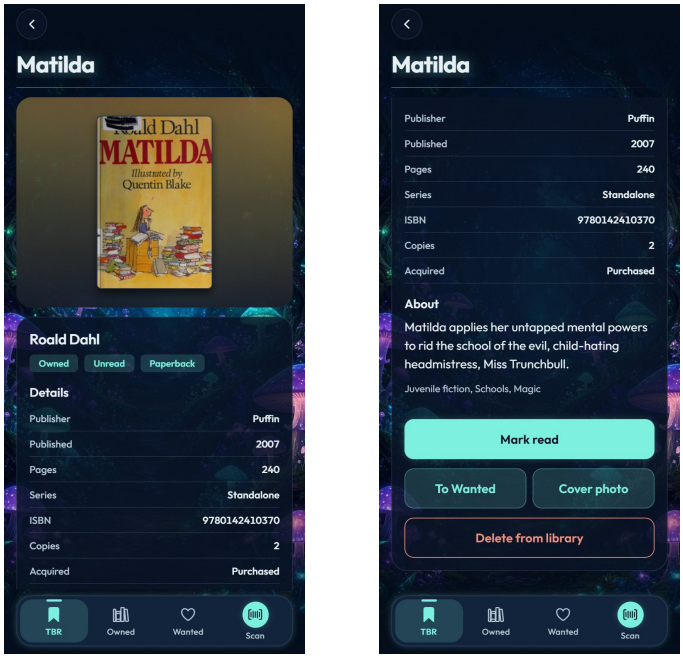


Figure 15: Book detail: cover and facts (left), description and actions (right)

Figure 15 shows a book's detail screen: the cover on a halo of its own colors, status tags, the edition's facts (publisher, date, pages, series, ISBN, copies, and how it was acquired), the description and subjects, and the actions: mark read or unread, move between shelves, replace the cover with a photo, and delete.

4 Detailed Design

Sequence Diagrams

Figure 16 shows the main use case. The lookup saves nothing; it returns the edition's details and cover candidates. Only after the owner confirms does `ScanBookUseCase` load the library, add a copy or a new `Book`, and save it.

Figure 17 shows how a domain rule is enforced. The view model gathers the acquisition first; `Book.moveTo` then refuses to move a book to Owned without a complete `Purchase` or `Gift`, and nothing is saved when the rule fails.

Figure 18 shows feature 4. After sign-in, `SyncLibraryUseCase` pulls the account's versions, keeps the newer copy of each book on both sides, and moves the books kept on the phone into the account. `AccountModel` then watches the account through Realtime: when another phone changes a book, this phone syncs and the shelves reload without a tap.

Pseudocode (scan a book)

```
User opens Scan
```

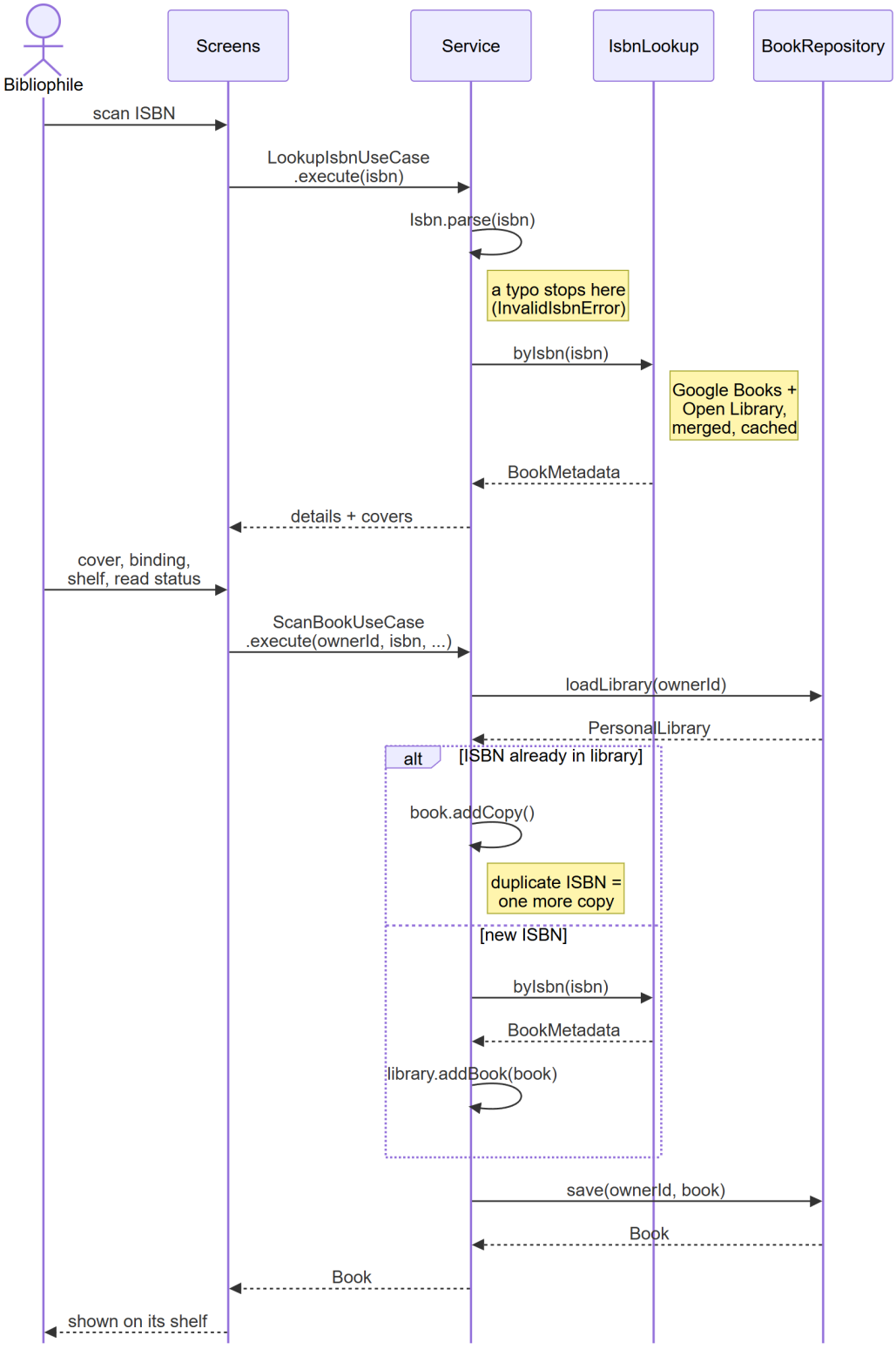


Figure 16: Sequence diagram: scanning a book (lookup, confirm, save)

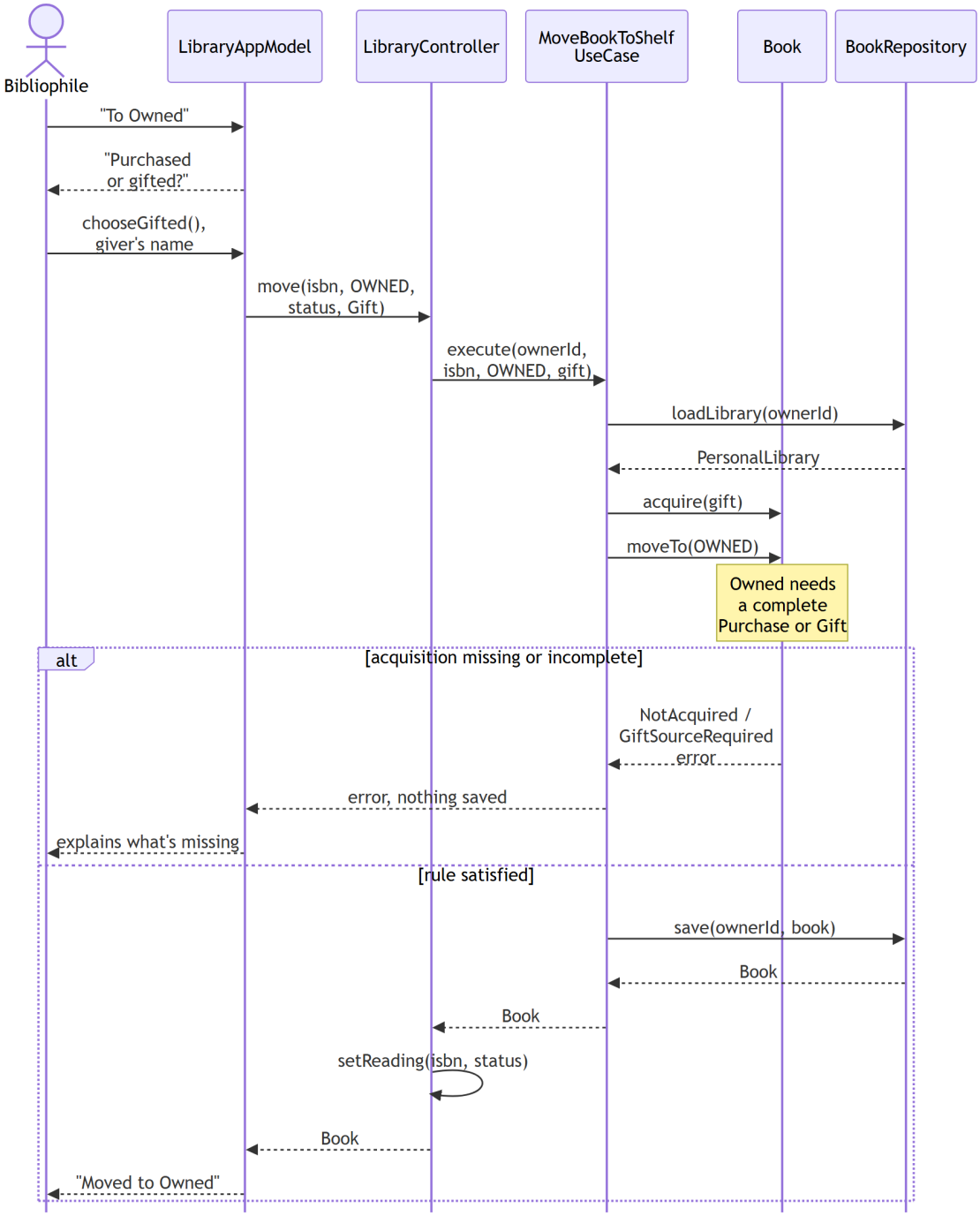


Figure 17: Sequence diagram: moving a Wanted book to Owned as a gift

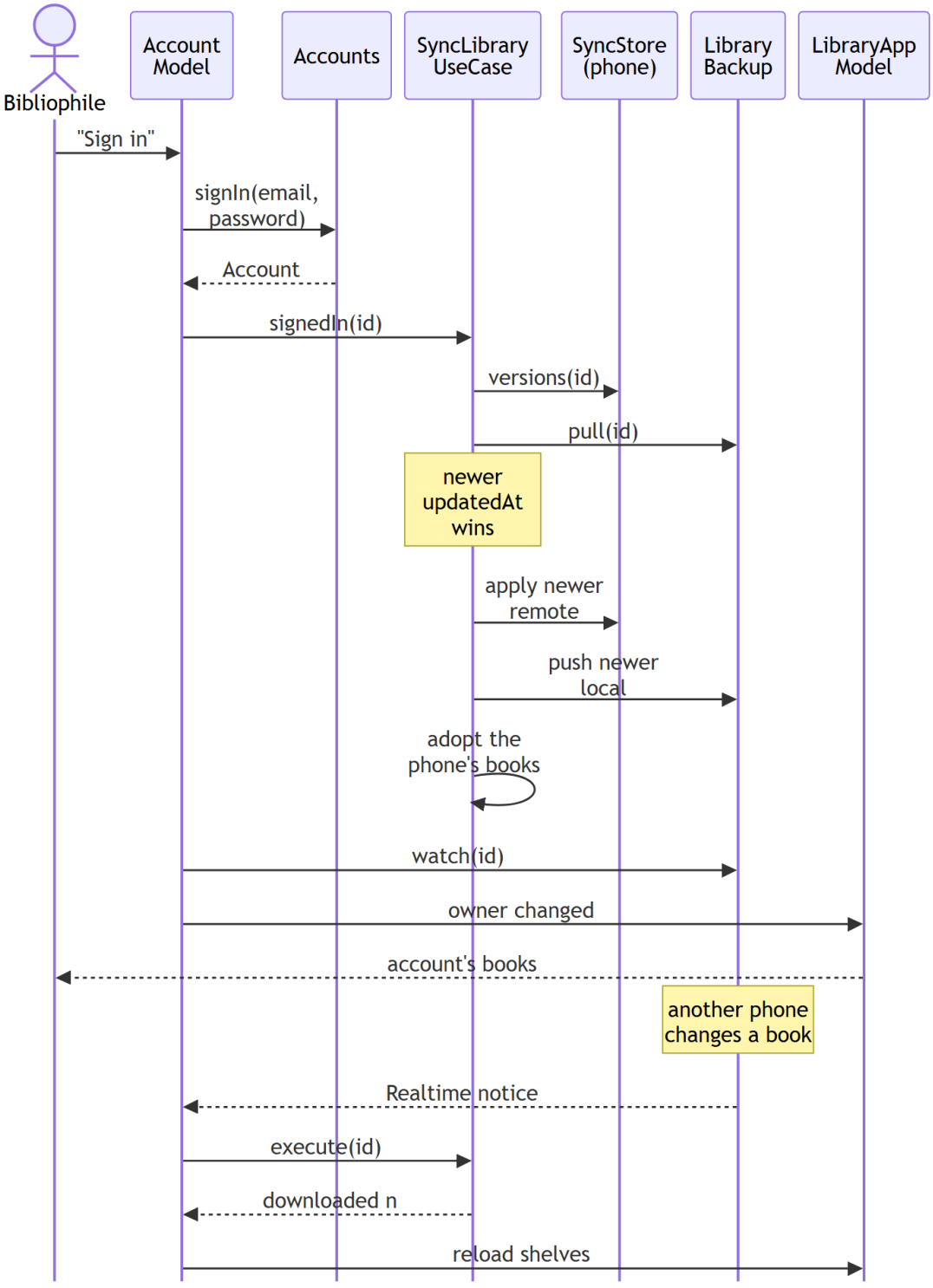


Figure 18: Sequence diagram: signing in, syncing, and auto sync (ADR-012)

```
If camera permission granted:
    Read the ISBN barcode from the camera preview
Else:
    User types the ISBN
metadata = cache.get(isbn) or merge(GoogleBooks(isbn), OpenLibrary(isbn))
If no source knows the ISBN:
    Show "No book found for ISBN ..."; stop
Show the edition and its cover candidates
If the ISBN is already in the library:
    On confirm: book.addCopy(); save(book); stop
Owner picks the cover that matches their copy (or photographs it)
If the binding is unknown:
    Ask "Hardcover or paperback?"
Ask "Owned or Wanted?"
If Owned:
    Ask "Purchased or gifted?"
    If gifted: ask for the giver's name (required)
Ask "Read or unread?"
library.addBook(new Book(metadata, choices))   // domain rules checked here
save(book); cache(metadata)
Show the book on TBR / Owned or Wanted
```

4.1 Data Detailed Design

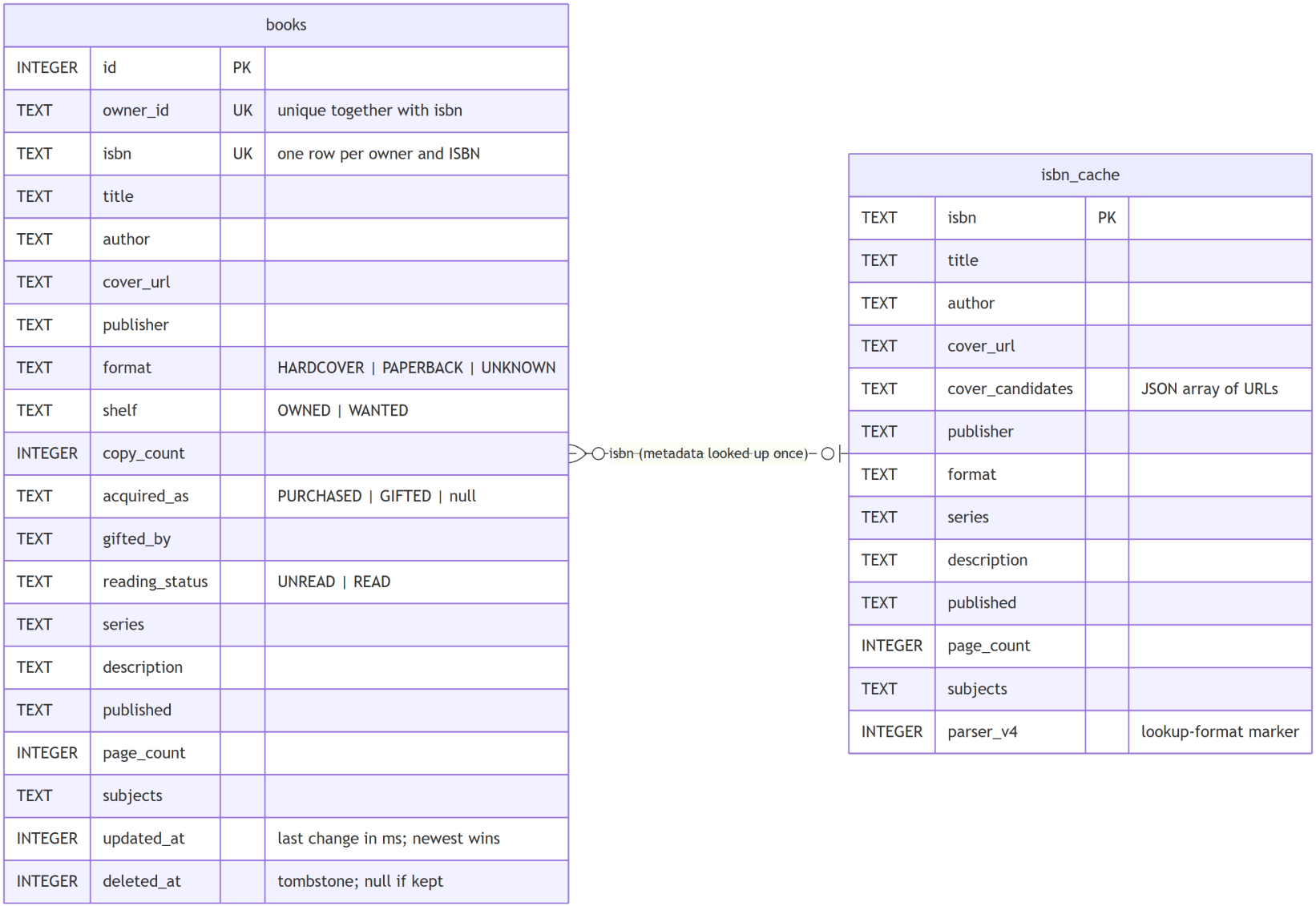


Figure 19: Local data model (SQLite)

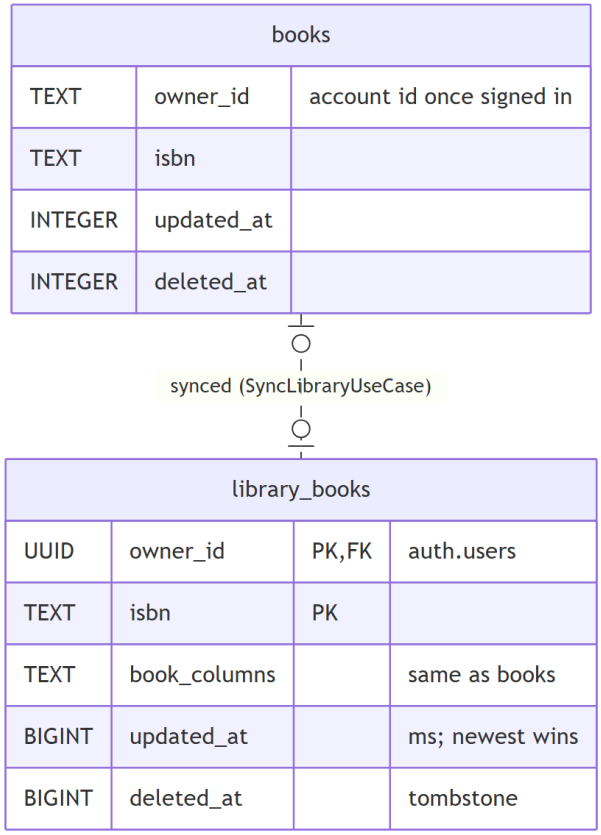


Figure 20: The synced copy: each account's `library_books` rows in Supabase, kept in step with the device's `books` rows

Figure 19 shows the two tables the app stores on the device, and Figure 20 the one table in Supabase. On the web the device data is kept as JSON in browser storage with the same field names.

- **books**: one row per cataloged title and owner: the edition facts (title, author, publisher, date, pages, series, subjects, description), the chosen cover (a URL or an inline photo), the binding, the shelf, the copy count, the reading status, and the acquisition (`PURCHASED` or `GIFTED` with the giver's name). Each owner has at most one row per ISBN (`UNIQUE (owner_id, isbn)`), which backs the "duplicate ISBN adds a copy" rule without letting one owner's books overwrite another's. Two sync columns record the last change (`updated_at`, in milliseconds) and the deletion (`deleted_at`): a deleted book stays as a tombstone, hidden from the shelves, so the deletion reaches other phones.
- **library_books** (Supabase): each account's synced copy, with the same book columns and the two sync columns, keyed by account and ISBN. Row level security lets a signed-in account read and write only rows whose `owner_id` is its own id, and only signed-in users are granted the table at all.
- **isbn_cache**: one row per looked-up ISBN with the merged metadata and every cover

candidate (as a JSON array), so a book is fetched from the network only once. A version marker clears old entries when the lookup format changes.

The schema migrates itself: missing columns are added when the app starts, and rows saved by older builds (for example the former third "TBR" shelf, now Owned and unread) are upgraded by `BookRowMapper`. The mapper also repairs damaged rows: a value outside its enumeration falls back to a safe default (an unknown shelf becomes Wanted, never Owned), and a row that no longer names its owner or ISBN is skipped. A browser save that is not readable JSON is kept under a backup key and the app starts with an empty library instead of failing. A database from the first release, where the ISBN was unique across all owners, is rebuilt once, inside a transaction, with the per-owner constraint.

4.2 RTM

The Requirements Traceability Matrix links each requirement to the components that implement it and to the verification cases in Section 7.

Req.-ID	**Requirement Description**	**Design Component**	**Test-Case #**
FR-1	Scan an ISBN with the camera, or type it	IsbnCamera, ScanForm, LibraryAppModel	15, 16, 21
FR-2	Look up the exact edition from Open Library and Google Books	OpenLibraryIsbnLookup, GoogleBooksIsbnLookup, MergedIsbnLookup	16, 21, 22
FR-3	Pick the matching cover or photograph your own	CoverPicker, ScanDraft, CoverCamera	17, 23
FR-4	Ask for the binding only when no source knows it	ScanDraft, LibraryAppModel	18
FR-5	Shelve a scanned book as Owned or Wanted	ScanBookUseCase, PersonalLibrary	14, 19
FR-6	An Owned book must be purchased or gifted; a gift needs the giver's name	Book, AcquisitionRecord, Purchase, Gift	10–14
FR-7	A duplicate ISBN adds a copy instead of a second entry	Book.addCopy, ScanBookUseCase	8
FR-8	Reading status is independent of the shelf; TBR shows Owned unread/read	Book, LibraryView, SetReadingStatusUseCase	4, 5, 24
FR-9	Browse Owned A–Z, by author, or by series with a letter rail	Catalog, CatalogGrouping strategies, AlphabetRail	6, 7
FR-10	Move a book between shelves	MoveBookToShelfUseCase, Book.moveTo	10–13

Req.-ID	Requirement Description	Design Component	Test-Case #
FR-11	Delete a book only after confirmation	DeleteBookUseCase, LibraryAppModel	25
FR-12	Cancel any popup without changing the library	LibraryAppModel .cancelPrompt	20
FR-13	Persist the library and cache lookups on the device	SqliteBookRepository, JsonBookRepository, CachingIsbnLookup	9
NFR-1	Tiers depend only downward; Service never imports Data	Ports, CompositionRoot, architecture tests	26
NFR-2	Stylesheets are imports; values come from theme tokens	presentation/styles, architecture tests	26
NFR-3	Runs in Expo Go (iOS, Android) and on the web without native builds	Expo modules only (ADR-005, ADR-008)	27
NFR-4	Accessible: WCAG AA contrast, screen-reader labels, reduced motion	Theme tokens, Tappable, Reveal, Labels	28
NFR-5	The architecture is documented: tier table, C4 and UML diagrams, ADRs	README, docs/adr, this SDD	1–3
FR-14	Sign in with an email and password (required to see or add books); the library is backed up to the account and kept in sync on every phone, automatically	AccountModel, AccountUseCases, SyncLibraryUseCase, SupabaseAccounts, SupabaseLibraryBackup, SyncStore	33–35
NFR-7	Each account reads and writes only its own books; only the public key ships in the app	Row level security on library_books, SupabaseConnection	33, 35
NFR-6	Bad input and failures are handled: a mistyped ISBN is refused, an unreachable source times out with a clear message, damaged storage still loads	Isbn, Http, BookSource-UnavailableError, BookRowMapper, JsonBookRepository, SqliteBookRepository	29–31

5 Sprint Planning

MyHomeLib is an individual project, so every development role below is held by the same developer, with a peer review partner reviewing the work at the midterm and final sessions (October 12 and 19, and November 9 and 16). Work is organized as weekly sprints that follow the course milestones. The *Proposed Timeline* below is the plan at project approval; the *Actual Timeline* records what was built. The sprint dates stayed in the same calendar windows; the content of Sprints 5 and 6 changed, as the Proposed Timeline table shows. Sprint 7 is in progress: alongside the midterm review it hardened the build against bad input and failures (ADR-010), verified in Section 7.7. The rest of the semester follows the Plan for the Second Half (Section 5.3), which freezes the scope to the four core features; Sprint 8 built feature 4 (Section 7.8).

5.1 Role Focuses

Team Member	**Role**	**Responsibilities**
Grace Hechavarria	Architect, Developer, UI/UX Designer, Tester	Requirements and domain rules, architecture and ADRs, all four tiers, interface design, automated tests, and documentation
Yousef Habouh	Peer Review Partner	Runs the app from the README, compares the diagrams to the code, and files the structured midterm and final reviews

5.2 Sprint Schedule, User Stories, and Acceptance Criteria

5.2.1 Proposed Timeline

The plan at project approval, one goal per weekly sprint, and what happened to it. The Actual Timeline below gives the user stories and acceptance criteria of the sprints that were built.

Sprint	Dates	Planned goal	What happened
1	08/24 – 08/30	Project proposal and approval	Done as planned.
2	08/31 – 09/06	Tier breakdown	Done as planned.
3	09/07 – 09/20	C4 and UML models	Done as planned (committed on 09/14).
4	09/21 – 09/27	Scanning, shelves, and copies	Done as planned.
5	09/28 – 10/04	Wanted list with a bookstore price check	Changed: scraping was removed at the instructor's request (ADR-004); the sprint built the acquisition rules instead.
6	10/05 – 10/11	Polish, tests, and midterm documentation	Changed: exact-edition lookup with Google Books (ADR-006), the class-based refactor and design system (ADR-007, ADR-008), tests, and this document.
7–9	10/12 – 11/22	Midterm review, device testing, final review and demo	Superseded by the Plan for the Second Half (Section 5.3), which freezes the scope to the four core features.

5.2.2 Actual Timeline

Sprint 1 – Project Proposal and Approval

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 1	08/24 – 08/30	As the instructor, I want a clear project idea and platform, so that I can approve the scope.	1. Verify that the repository exists with a README. 2. Verify that the approval paragraph names the platform, features, and the one external integration.

Team Member	Task
Grace Hechavarria	Created the repository; drafted three project ideas; wrote and updated the approval paragraph (PR #1)

Sprint 2 – Tier Breakdown

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 2	08/31 – 09/06	As a developer, I want each tier's responsibilities written down, so that every class has an obvious home.	1. Verify that the README has a Presentation / Service / Domain / Data table. 2. Verify that each row names the system's real responsibilities, not generic ones.

Team Member	Task
Grace Hechavarria	Wrote the tier responsibilities table (PR #2)

Sprint 3 – C4 and UML Models

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 3	09/07 – 09/20	As a reviewer, I want C4 and class diagrams, so that I can understand the system before reading code.	1. Verify that the README has C4 Context, Container, and Component diagrams. 2. Verify that a UML class diagram shows the domain rules. 3. Verify that the diagrams render on GitHub (Mermaid).

Team Member	Task
Grace Hechavarria	Drew the C4 Level 1–3 diagrams and the UML class diagram in Mermaid

Sprint 4 – Initial Architecture and Core Logic (Issue #3)

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 4	09/21 – 09/27	As a bibliophile, I want to scan or type an ISBN and see the book's details, so that I can add it without typing.	1. Verify that the camera reads an ISBN barcode. 2. Verify that a typed ISBN works when there is no camera. 3. Verify that the lookup shows title, author, and cover, and saves nothing.
		As a bibliophile, I want a book I already own to count as another copy, so that my catalog has no duplicate entries.	1. Verify that scanning a cataloged ISBN increases the copy count. 2. Verify that no second entry is created.
		As a bibliophile, I want separate Owned and Wanted shelves and a To-be-read list, so that I know what to read and what to buy.	1. Verify that a book is either Owned or Wanted. 2. Verify that TBR lists Owned books, split Unread / Read. 3. Verify that the library survives an app restart.

Team Member	Task
Grace Hechavarria	Built the four-tier skeleton, the domain rules, the scan flow (camera and typed ISBN), the Open Library adapter, and the SQLite and JSON repositories (PR #4)

Sprint 5 – Acquisition Rules (Issue #5)

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 5	09/28 – 10/04	As a bibliophile, I want every Owned book to record whether I bought it or received it as a gift, so that I remember where each book came from.	1. Verify that a Wanted book cannot move to Owned without a purchase or a gift. 2. Verify that a gift requires the giver's name. 3. Verify that scanning straight into Owned asks the same question. 4. Verify that a refused move saves nothing.

Team Member	Task
Grace Hechavarria	Enforced the acquisition rules in the domain model and every path into Owned, with tests (PR #6); the bookstore price check was dropped (ADR-004)

Sprint 6 – Exact Edition, Class-Based Refactor, and Design System

Sprint	Dates	User Stories	Acceptance Criteria
Sprint 6	10/05 – 10/11	As a bibliophile, I want the app to find the exact edition I am holding, so that the publisher, binding, and cover match my copy.	1. Verify that Google Books and Open Library are both asked and merged. 2. Verify that every distinct cover is offered, plus "Your photo". 3. Verify that the binding is asked only when no source knows it.
		As a developer, I want the code built from classes with no duplication and imported stylesheets, so that it is modular and easy to review.	1. Verify that screen logic lives in a view model class, not in components. 2. Verify that no component defines a stylesheet or inline style. 3. Verify that the tier rules hold (architecture tests pass).
		As a bibliophile, I want a polished, consistent interface, so that the app is pleasant to use every day.	1. Verify that every screen uses the same tokens, font, and building blocks. 2. Verify that text meets WCAG AA contrast. 3. Verify that motion stops when Reduce Motion is on.

Team Member	Task
Grace Hechavarria	Added the Google Books adapter, merged and cached lookups, the cover picker and own-cover photo, and the binding question; refactored the presentation into an MVVM view model; built the design system; brought the suite to 88 automated tests; wrote ADRs 006–009 and this SDD

5.3 Plan for the Second Half

The scope for the final is frozen to the four core features (Section 1.1): **nothing new is added until these four are complete and tested**. The second half completes and tests features 1–3, builds and tests feature 4, verifies all four on physical phones, and closes with the final review. This plan supersedes Sprints 7–9 of the Proposed Timeline. The work is tracked as GitHub issues in the title / why / "done when" form and shown on the project board.

Weeks	**GitHub issue**	**Done when**
8–9 (10/12 – 10/25)	Complete and test features 1–3; address midterm findings	Every rule of features 1–3 has a passing test (`tests/coreFeatures.test.ts` and the unit tests), every midterm peer-review finding and the check-in action item is fixed or answered, and CI is green.
10–11 (10/26 – 11/08)	Account library sync (feature 4)	Done: ADR-012 records the approach (email accounts through Supabase instead of Google sign-in); the owner signs in, and the library is backed up to the account and kept in sync on another phone; each account's books stay separate (contract and sync tests); CI is green.
12 (11/09 – 11/15)	Verify all four features on a physical phone	Each core feature is walked through in Expo Go on an iPhone and an Android phone, defects are fixed, and the results are recorded.
13 (11/16 – 11/22)	Final review, demo, and SDD v2.0	The final peer review is addressed, the demo runs from a clean clone, and this document is updated to its final version.

Risk	Mitigation
A book source is slow or down (Open Library took about 20 seconds per request during testing)	Every lookup is cached on the device after the first time; Google Books answers when Open Library does not; a typed ISBN and a clear "No book found" message cover the rest; the demo uses books already cached.
The camera, haptics, and SQLite have only been exercised in the web build so far	Week 12 is reserved for testing all four features on physical iPhone and Android phones in Expo Go, before the final review; SQLite is already tested against a real database in CI.
Google sign-in may not work inside Expo Go (Google's native sign-in library needs a custom development build, and Google restricts which redirect addresses it accepts)	Happened: a custom build is not free on iPhone, so ADR-012 chose email accounts through Supabase, which run in Expo Go on both phones.
Account sync could be read as a second data store, against the one-store course constraint	Recorded in ADR-012 as the developer's decision: SQLite stays the only working store, and the account copy is a synced copy behind its own ports.
Sync grows beyond its feature (conflict handling, sharing, several devices at once)	Feature 4 is limited to sign-in and sync of one owner's library; a later change of the same book wins (by the phones' clocks). Password reset and sharing wait until after the final.
An Expo or dependency update breaks the app before the demo	Happened in week 11: Expo Go moved to SDK 58 and would no longer open an SDK 57 project, so the project was upgraded the same day (`expo-doctor` clean, every test passing). Versions stay pinned in `package-lock.json`, and CI installs with `npm ci` on every push.
The Google Books key is missing or over quota	The key is optional: without it the app uses Open Library only (ADR-006), and the key is restricted to the Books API.
Review findings take longer than planned	Weeks 8–9 hold no new features, only findings and tests for features 1–3, ordered by how much they affect the grade and the user.
One developer, limited time	Work is split into small issues with weekly commits, and CI and the 195 automated tests catch regressions immediately.

6 Design

6.1 Definition of Design

Design in software engineering converts requirements into a blueprint for the system: its architecture, its main elements, and how they interconnect. The design phase decides what the elements are and how they communicate before (and while) they are built and tested. Sommerville (2021) describes software design as the process of identifying system components and their relationships so that the system can be implemented and verified.

6.2 Design Process

The design followed the course sequence. First, the problem and constraints were written as an approval paragraph. Next, the responsibilities of each tier were listed. Then the system was modeled at increasing zoom with the C4 model (Context, Container, Component) and UML class and sequence diagrams, and each significant decision was captured in an ADR.

The design was centered on the core workflow, scanning a book:

1. The user scans or types an ISBN.
2. The system looks up the exact edition in its book sources.
3. The user confirms the edition and picks the cover that matches their copy.
4. The user chooses a shelf, how the book was acquired (for Owned), and whether it has been read.
5. The system applies the domain rules and saves the book.

The domain rules were modeled as behavior of the domain classes rather than checks in the screens, so they hold no matter which screen triggers them. After the midterm build worked, the code was refactored toward the object-oriented techniques required by the course (abstract base classes, interfaces, polymorphism, composition, and design patterns), and the diagrams were regenerated to match the code.

6.3 Graphical Representation of the Design

6.3.1 Use Case Diagram

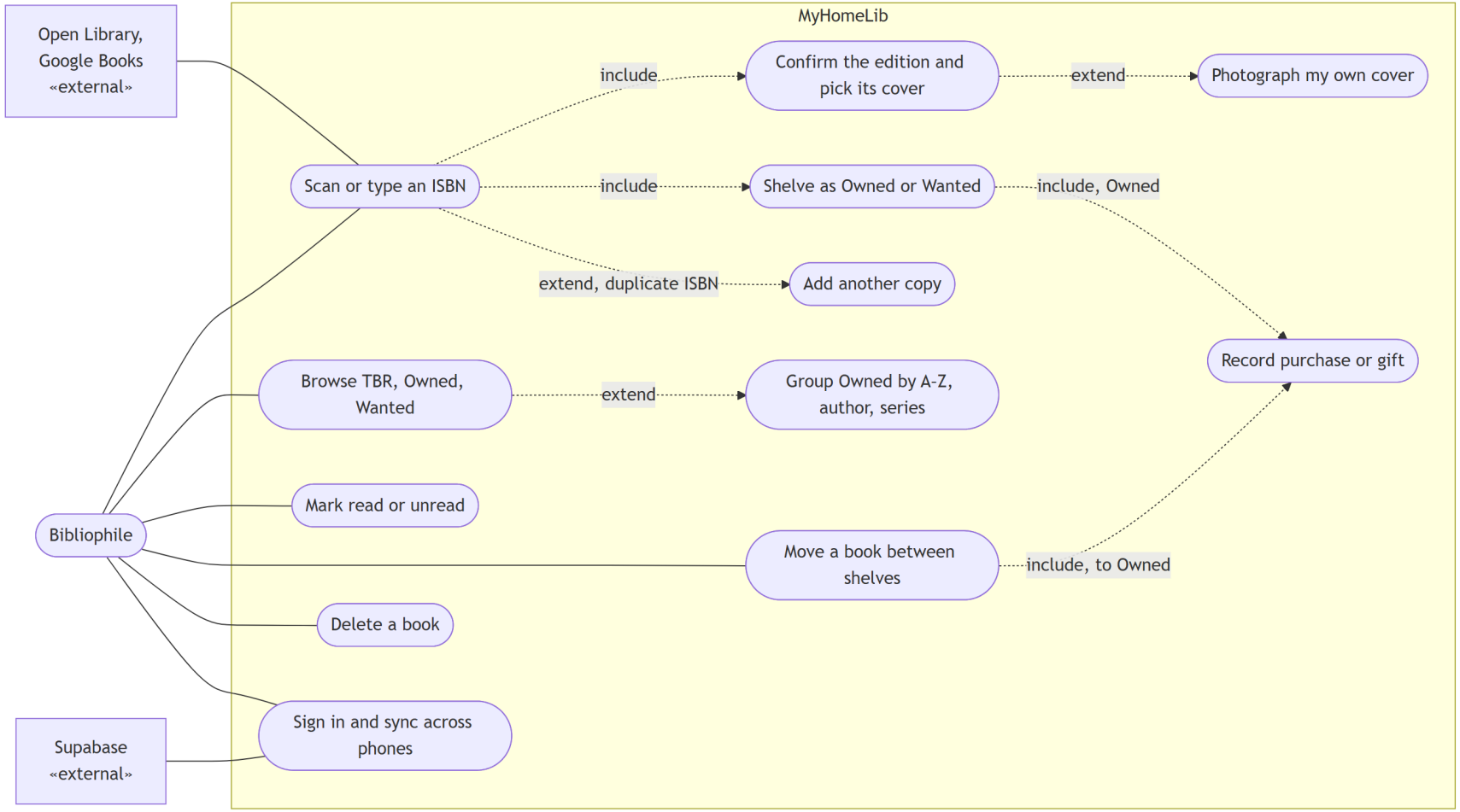


Figure 21: Use Case Diagram

Figure 21 shows what the bibliophile can do. Scanning *includes* confirming the edition and choosing a shelf; choosing Owned *includes* recording a purchase or gift. A duplicate ISBN *extends* scanning into adding a copy. The two book sources are external actors of the scan use case. Signing in *includes* syncing the library, with Supabase as its external actor.

6.3.2 Activity Diagram

Figure 22 shows every decision in the scan flow, including the typed-ISBN fallback, the unknown-ISBN exit, the duplicate-copy shortcut, the optional binding question, and the gift branch.

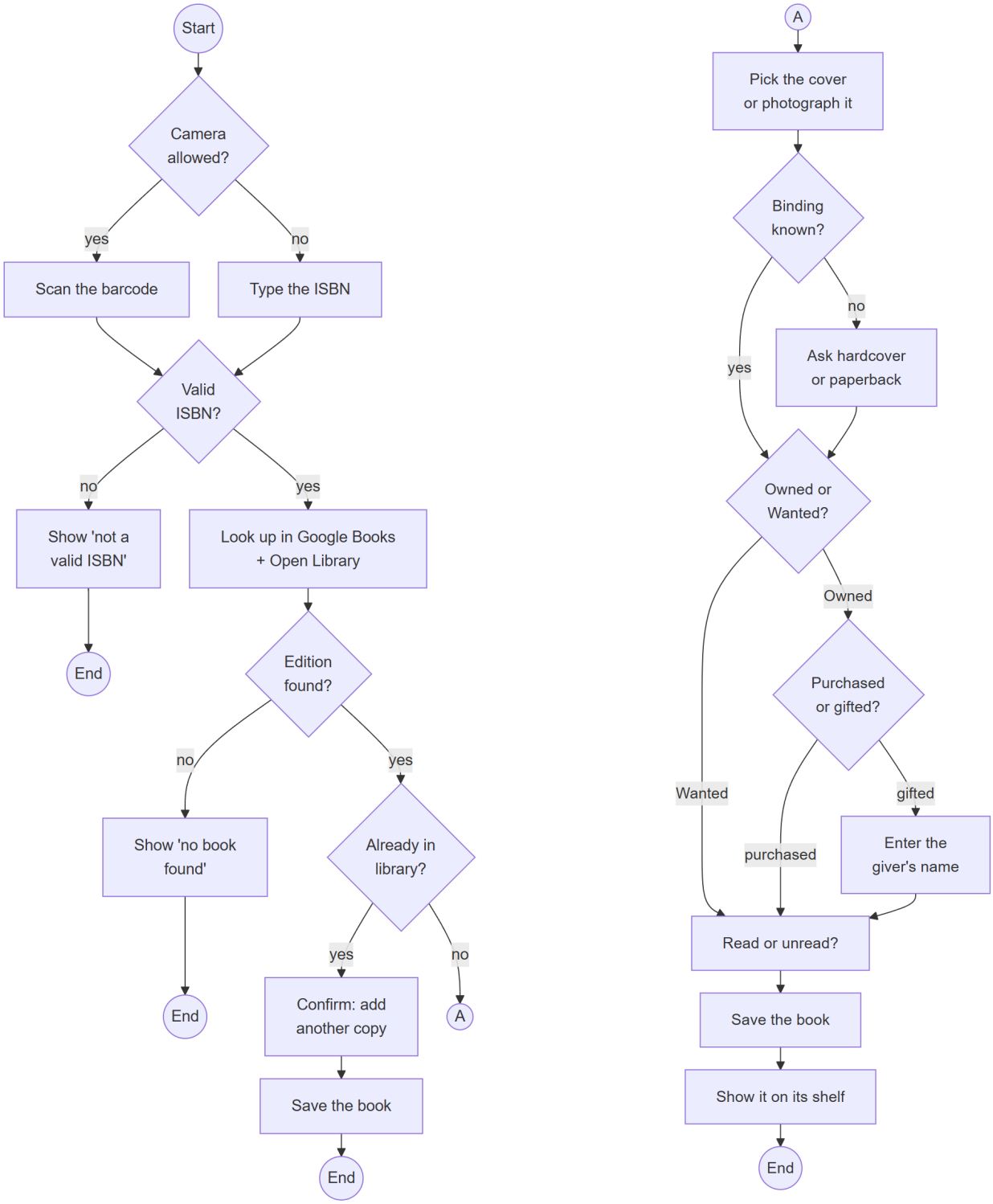


Figure 22: Activity Diagram: scanning a book. Left: finding the edition; right: shelving a new book, continuing at A

6.3.3 Class Diagram

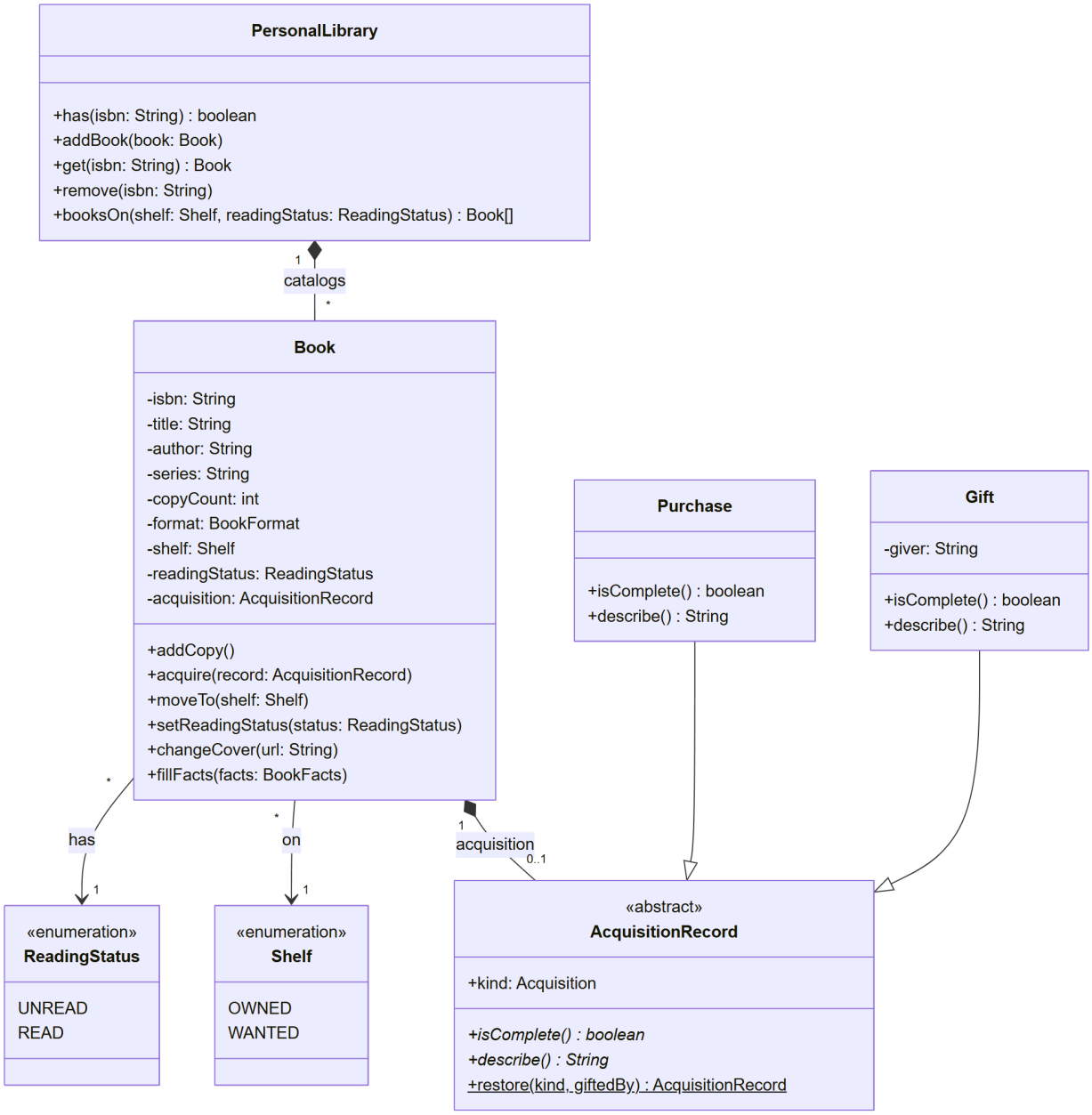


Figure 23: Class Diagram: Domain model

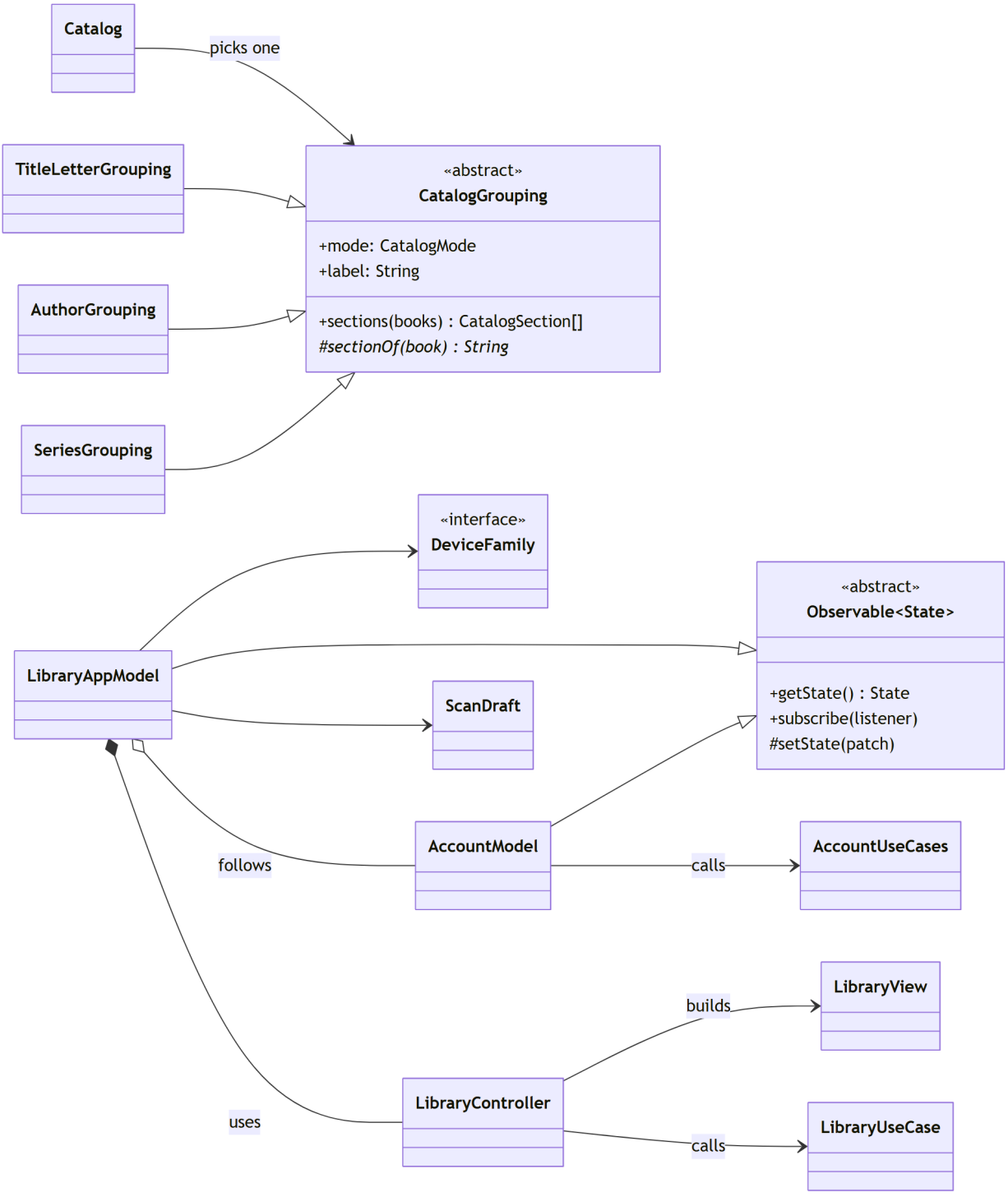


Figure 24: Class Diagram: Presentation view model and catalog strategies

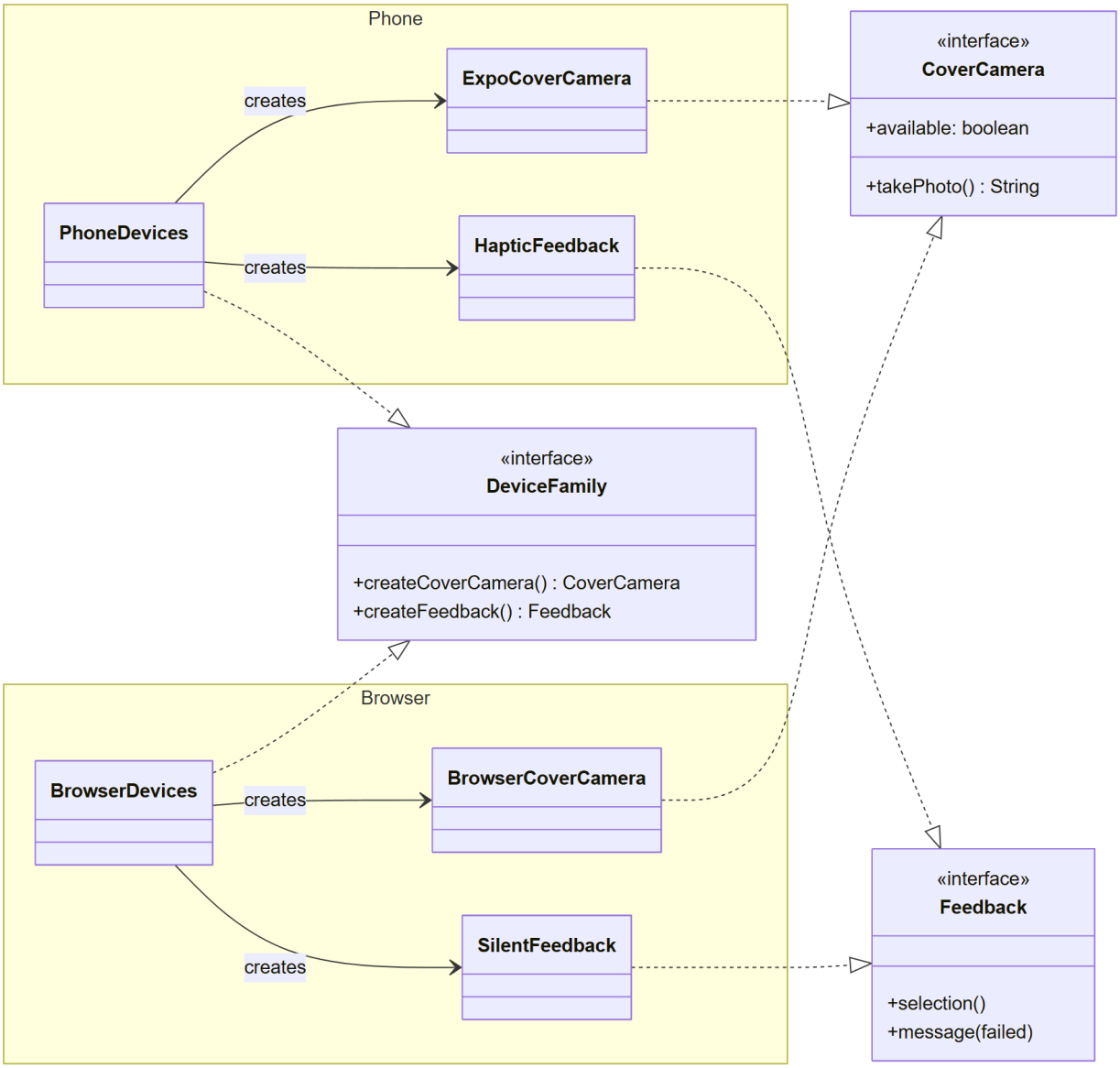


Figure 25: Class Diagram: the device families (Abstract Factory, ADR-011)

Figure 23 is the core class diagram: the five domain classes that carry the system's rules (`PersonalLibrary`, `Book`, `AcquisitionRecord`, `Purchase`, `Gift`) with their real attributes and relationships, plus two enumerations. The other tiers' classes are drawn where their interfaces are described, so they are not repeated here: Service in Figure 6, Data in Figures 8 and 9, and accounts and sync in Figure 10. Figure 24 adds the Presentation view model and the catalog strategies, and Figure 25 the device families. The object-oriented techniques used, and where:

Technique	Where it is used
Encapsulation	`Book` keeps its state private; it changes only through rule-checking methods.

Technique	Where it is used
Abstract class + inheritance	`LibraryUseCase` → `LibraryLookupUseCase` → use cases; `AcquisitionRecord`; `CatalogGrouping`; `Observable`.
Polymorphism	`Purchase` and `Gift` answer `isComplete()` and `describe()` their own way.
Interfaces	`BookRepository`, `IsbnLookup`, `IsbnCache`, `SyncStore`, `LibraryBackup`, `Accounts`, `CoverCamera`, `Feedback`, `DeviceFamily`.
Composition	`PersonalLibrary` composes `Books`; `JsonBookRepository` composes `MemoryBookRepository`.
Template Method	`LibraryUseCase.updateBook` (load → change → save).
Strategy	`TitleLetterGrouping`, `AuthorGrouping`, `SeriesGrouping`.
Decorator	`CachingIsbnLookup` and `MergedIsbnLookup` wrap other lookups.
Abstract Factory	`DeviceFamily`: `PhoneDevices` or `BrowserDevices` creates a matched camera and feedback (the other four creational patterns are checked in `docs/pattern-map.md`).
Simple Factory	`UseCases` and `AccountUseCases` build every use case; `AcquisitionRecord.restore`; `SqliteBookRepository.open`; `SupabaseConnection.open`.
Null Object	`UnconfiguredAccounts` and `UnconfiguredLibraryBackup` stand in when no Supabase project is configured, so no caller checks for one.
Data Mapper	`BookRowMapper` is the only row ↔ `Book` translation.
Observer / MVVM	`LibraryAppModel` and `AccountModel` extend `Observable`; React subscribes with `useObservable`; Realtime change notices start a sync.
Dependency injection	`CompositionRoot` passes adapters to the use cases; `App.tsx` passes a device family to the view model; tests pass fakes.

6.3.4 Sequence Diagram

The sequence diagrams for scanning a book, for moving a Wanted book to Owned, and for signing in and syncing are shown in Figures 16, 17 and 18 (Section 4).

6.4 Rationale Management

Rationale management records why each design decision was made. In this project every significant decision is recorded as an ADR in `docs/adr/`, so the reasoning stays next to the code and can be reviewed together with it.

ADR	Decision	Status
001	TypeScript + Expo Go modular monolith with four tier folders	Accepted (external-source line amended by 006)
002	SQLite on the device, JSON fallback on the web, one port	Accepted
003	Open Library for ISBN metadata (no key)	Accepted, amended by 006
004	No bookstore scraping	Accepted
005	Expo Go instead of an app-store binary	Accepted
006	Google Books as a second book source (optional key)	Accepted
007	MVVM presentation with an observable view model	Accepted
008	Design system: tokens, stylesheet imports, shared UI building blocks	Accepted
009	Ports and adapters between Service and Data	Accepted
010	Validate input and contain failures at the tier edges	Accepted
011	Abstract Factory for the device services	Accepted
012	Account sign-in and library sync with Supabase (email sign-up)	Accepted (developer's decision)

6.4.1 Issues Addressed

The first issue was **portability**: the review partner had to run the app without a Mac or paid accounts, which ruled out native builds and led to Expo Go. The second issue was **trustworthy data**: a catalog is only useful if it cannot contain duplicate entries or Owned books with no record of how they were acquired, so those rules had to live in the domain model where no screen can bypass them. The third issue was **matching the physical copy**: early testing showed that one ISBN can map to a wrong or missing cover, so the lookup had to become edition-first, consult a second source, and let the owner choose. The fourth issue was **keeping the code reviewable**: logic that had grown inside one large screen component had to move into classes that can be tested and diagrammed.

6.4.2 Alternatives Considered

- **Native iOS app (Swift/Xcode)** or a **Flask web app**: rejected; the first needs a Mac, and the second is not a mobile app (ADR-001, ADR-005).
- **Scraping bookstore prices for the Wanted list**: rejected for reliability and terms-of-service reasons (ADR-004).
- **Open Library only**: simpler, but often wrong for a specific printing; Google Books was added as an optional second source (ADR-006).
- **Keeping screen logic in React hooks**: faster to write, but untestable without rendering and not class-based; replaced by an MVVM view model (ADR-007).
- **A UI component library**: would add dependencies and a generic look; a small token-based design system was built instead (ADR-008).
- **Google sign-in for feature 4**: rejected for now; it needs a custom native build, which is not free on iPhone, while email sign-in through Supabase runs in Expo Go on both phones (ADR-012). **Writing our own sign-in server** was rejected too: password storage, reset emails and hosting are more work and more risk.
- **Plain JavaScript instead of TypeScript**: JavaScript is required by React Native, but it has no interfaces, abstract classes, or access modifiers, and it reports type mistakes only at run time. TypeScript was chosen because those features carry the course's object-oriented design straight into the code (the UML diagrams match the classes), the compiler refuses an adapter that does not implement its port, and all four tiers and the tests share one language.
- **Microservices or a backend server**: rejected; the library is private to one user and must work offline on a phone, so a server would add cost and failure points without benefit. Account sync (feature 4) uses Supabase, a hosted service, through its official client, so the project still runs no server of its own (ADR-012).

6.4.3 Decisions Made

The design is centered on the scan workflow and the collection rules. The rules are methods of the domain classes; the use cases only orchestrate load, change, and save; the Data tier is replaceable behind its ports (library, ISBN cache, and book sources, each kept to what its callers use); and the screens are thin views of one observable view model. External sources are combined behind a single interface, so adding or removing a source does not change any caller.

6.4.4 Criteria Used to Guide Decisions

- The design must satisfy the course constraints (portable platform, one store, no scraping, documented architecture).
- The diagrams must match the code, and automated tests must keep it that way.
- Each class must have one responsibility, and no logic may be written twice.
- The design must stay simple enough to finish in one semester and explain in a review.

6.4.5 Design Debate and Final Resolution

The main debate was between keeping the external integrations to the minimum (Open Library only) and matching the user's exact copy. Feedback from the instructor favored limiting integrations and removing scraping, while testing showed that a single free source produced wrong covers for many editions. The resolution was to add Google Books only through its official API, make its key optional so the app still runs without it, merge both sources behind the existing `IsbnLookup` port, and always let the owner choose or photograph the cover. This kept the architecture unchanged, respected the no-scraping rule, and met the user's need. A second debate concerned how much structure to add for the course's object-oriented requirements; the resolution was to introduce each pattern only where it removed real duplication or made a rule testable, and to document each one in the table above. A third debate concerned feature 4: the approved scope said "Google Account", but Google sign-in could not run free on both phones in Expo Go. The resolution kept the feature's goal (one library on every phone) and changed the sign-in method, recorded in ADR-012 with the scope wording updated everywhere.

7 Verification

Verification in software is defined by the IEEE 1012 standard as "the process of evaluating software to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase" (IEEE, 2024).

MyHomeLib was verified in two complementary ways. First, **automated tests** (Jest) exercise every tier: 195 tests in 15 files, including one acceptance test file that drives each core feature end to end, including tests that walk the view model through each user flow without any user interface, persistence tests against a real SQLite database, tests that feed bad input and failing networks on purpose, and tests that enforce the architecture rules. Second, each user story was **walked through in the running application**, and the screenshots below show the results. The numbered cases match the Test-Case column of the RTM.

Test file	What it verifies	Tests
domain.test.ts	Book and library rules: copies, acquisition, shelves, reading status	17
useCases.test.ts	Every use case against in-memory adapters	23
bookSources.test.ts	Open Library and Google Books parsing, merging, failures, caching	16
appModel.test.ts	Every user flow through the view model, no UI	17
controller.test.ts	Controller, library view, JSON round trip	6
catalog.test.ts	A–Z, author, and series grouping	3
persistence.test.ts	BookRowMapper (legacy and damaged rows), the repository, cache and sync contracts for memory, JSON and SQLite, owner isolation, tombstones, schema upgrades	42
resilience.test.ts	ISBN check digits and ISBN-10 to ISBN-13 keys, timeouts, error statuses, malformed JSON, the banner shown for each	17
coreFeatures.test.ts	Each of the four core features end to end, with the real lookup pipeline on canned JSON; a library-only store runs every use case	12
devices.test.ts	The device family: the browser family as a whole, feedback through the family, camera availability	5
accounts.test.ts	Credentials and the sign-up, sign-in and sign-out use cases	4
supabaseAdapters.test.ts	Supabase sign-in errors in plain words, the Null Objects, backup round trip, paging, Realtime watch, offline	11
sync.test.ts	Two phones: merge, latest change wins, deletes stay deleted, offline, first sign-in, isolation	7
accountModel.test.ts	The Account tab with the library: sign-in lock, sign up, wrong password, sign out, auto sync, offline, no service	9
architecture.test.ts	Tier dependency rules and stylesheet rules	6
Total	**Test Suites: 15 passed; Tests: 195 passed**	**195**

Table 3: Automated test suite results (`npm test`, November 7, 2026)

7.1 Sprint 1 – Project Proposal and Approval

Process	Verification	Figure
(1) We open the repository's README. It contains the approval paragraph, which names the platform (Expo Go on iOS and Android, with Expo web as a fallback), the core features, and the external book source.	Verify that the approval paragraph names the platform, features, and integration	README

7.2 Sprint 2 – Tier Breakdown

Process	Verification	Figure
(2) We read the README's tier table. Each of Presentation, Service, Domain, and Data lists this system's own responsibilities, and each matches a top-level folder in the code.	Verify that each tier's real responsibilities are written down	Fig. 2

7.3 Sprint 3 – C4 and UML Models

Process	Verification	Figure
(3) We open the README on GitHub. The C4 Context, Container, and Component diagrams and the UML class and sequence diagrams render from Mermaid, and the class names match the code (the architecture tests keep the tiers as drawn).	Verify that the C4 and UML diagrams exist, render, and match the code	Fig. 1, Fig. 23

7.4 Sprint 4 – Scanning, Shelves, and Copies

Process	Verification	Figure
(4) We open the app on the To be read tab with a small library. Owned unread books are listed; the switch shows the unread and read counts.	Verify that TBR lists Owned books split Unread / Read	Fig. 11
(5) We switch to Read. Only finished Owned books appear, marked with a leaf. With an empty library the shelf explains itself and offers Scan.	Verify that reading status is independent of the shelf	Fig. 11
(6) We open Owned and switch between A–Z, Author, and Series. Each grouping lists the same books under index headings.	Verify the three catalog groupings	Fig. 12
(7) In Series, the Dune series appears as a fan of covers with a count badge of 2; tapping it lists both books.	Verify series grouping and expansion	Fig. 12
(8) We scan the ISBN of Matilda, which is already cataloged. The popup says it is already in the library; confirming adds a copy, and the banner reports "Copies: 2" without a second entry.	Verify that a duplicate ISBN increments the copy count	Fig. 26
(9) Repository tests save a library through browser (JSON) storage, reload it, and get the same books back, including a gift restored as a `Gift`; a cache test looks up the same ISBN twice and the book sources are asked only once.	Verify that the library persists and lookups are cached	Table 3

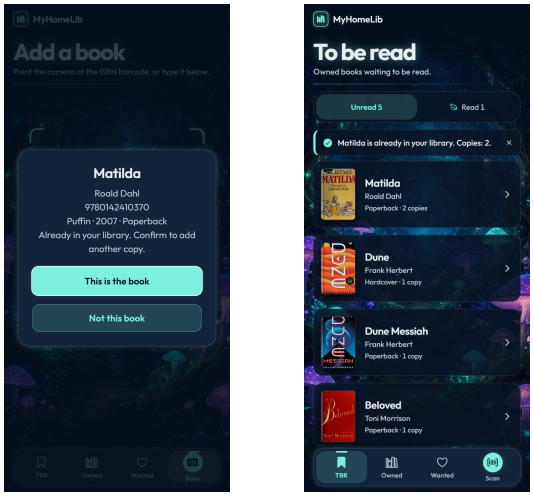


Figure 26: A duplicate ISBN offers to add a copy (left); the copy count becomes 2 (right)

7.5 Sprint 5 – Acquisition Rules

Process	Verification	Figure
(10) We open Beloved on the Wanted shelf and tap "To Owned". Before anything moves, the app asks whether it was purchased or gifted.	Verify that a Wanted book cannot move to Owned without a purchase or gift	Fig. 27
(11) We choose Gifted and tap Continue without a name. The popup stays open and nothing changes.	Verify that a gift requires the giver's name	Fig. 27
(12) We type the giver's name, continue, and answer Unread. The banner reports "Moved to Owned (unread)" and the book appears in Owned.	Verify that a complete gift moves the book	Fig. 28
(13) Domain tests try every move without an acquisition, with an empty giver, and with a blank giver; each is refused and the saved book stays Wanted.	Verify that a refused move saves nothing	Table 3
(14) While scanning a new book we choose Owned; the app asks "How did you get it?" before the reading question.	Verify that scanning into Owned asks for the acquisition	Fig. 30

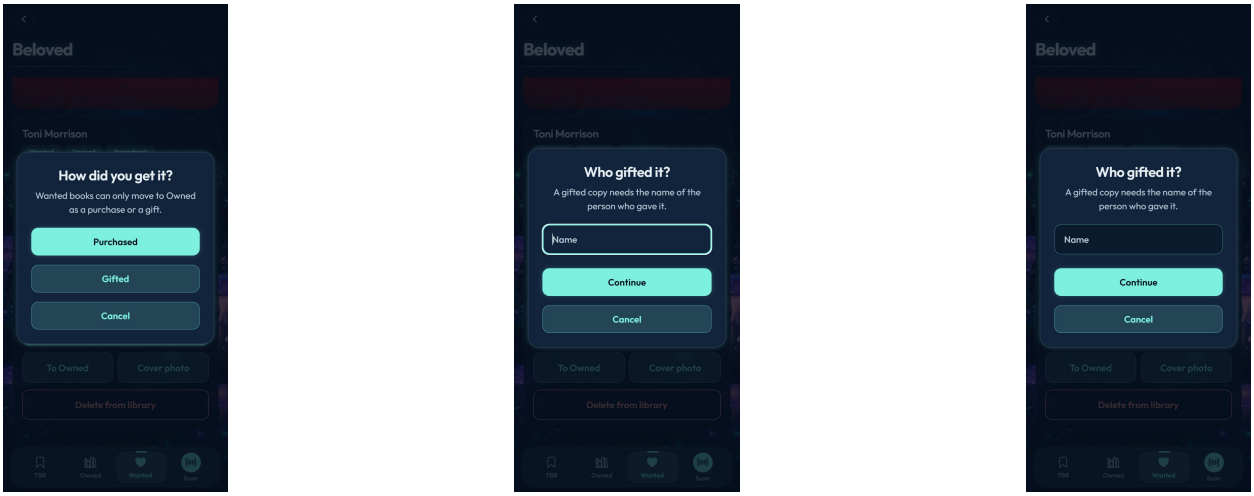


Figure 27: Moving to Owned asks for the acquisition (left); a gift needs the giver's name (center, right)

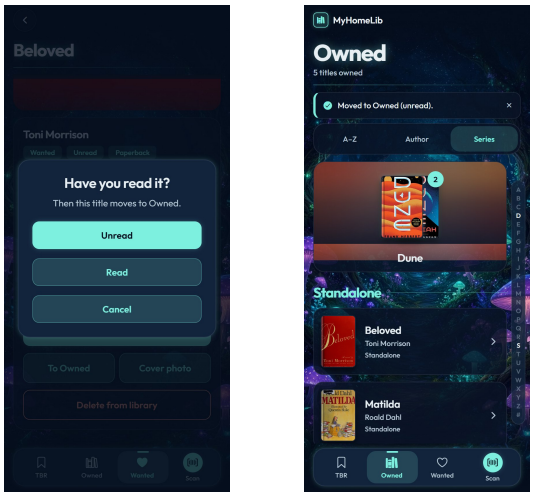


Figure 28: The last question is read or unread (left); the book moves to Owned (right)

7.6 Sprint 6 – Exact Edition, Refactor, and Design System

Process	Verification	Figure
(15) We open Scan. The camera preview shows the barcode target, and a typed-ISBN field is always available.	Verify the camera and the typed fallback	Fig. 29
(16) We type the ISBN of The Great Gatsby and look it up. The popup shows the edition (publisher, year) and saves nothing yet.	Verify that the lookup returns the edition and does not save	Fig. 29

Process	Verification	Figure
(17) The popup offers every distinct cover the sources returned plus "Your photo". We tap the second cover; the check badge and the halo follow the choice.	Verify the cover picker	Fig. 29
(18) The sources did not state a binding, so the app asks "Hardcover or paperback?". A book whose binding is known skips this step (automated test).	Verify that the binding is asked only when unknown	Fig. 30
(19) We choose Owned, Purchased, and Unread. The banner reports "Added The Great Gatsby to Owned (unread)".	Verify that the scanned book is saved with every choice	Fig. 31
(20) We start another scan and tap "Not this book". The banner reports "Nothing added" and the library is unchanged.	Verify that cancelling a popup changes nothing	Fig. 31
(21) We look up an ISBN that no source knows. The banner reports "No book found for ISBN . . . ".	Verify the unknown-ISBN message	Fig. 32
(22) Book-source tests feed recorded responses: edition data wins over work data, a different edition is never passed off as this one, every distinct cover is kept, and one source failing is tolerated.	Verify the edition-first merge	Table 3
(23) View-model tests photograph a cover with a fake camera and check that the photo is saved; a refused camera explains itself and changes nothing.	Verify "Your photo" and camera refusal	Table 3
(24) On a book's detail screen we tap "Mark read"; the banner reports "Marked read." and the book moves to the Read list.	Verify the read toggle	Fig. 32
(25) We tap "Delete from library"; a confirmation names the book and warns that every copy is removed. "Keep it" leaves it in place.	Verify that delete asks first	Fig. 32

Process	Verification	Figure
(26) The architecture tests read every source file: tiers depend only downward, no component defines a stylesheet or inline style, and only `App.tsx` reaches the composition root. All 6 pass.	Verify the tier and stylesheet rules	Table 3
(27) We start the project with `npx expo start` and open it in Expo Go on a phone; no native build tools are required. In a web browser the app also runs, with the camera replaced by a message to use the phone and a typed ISBN.	Verify the platform constraint	Fig. 11
(28) Every text and background color pair was measured with the WCAG formula (lowest: 4.63:1, the danger button); cards announce "title, by author, . . . , read" to screen readers; list entrances stop when Reduce Motion is on.	Verify accessibility	–

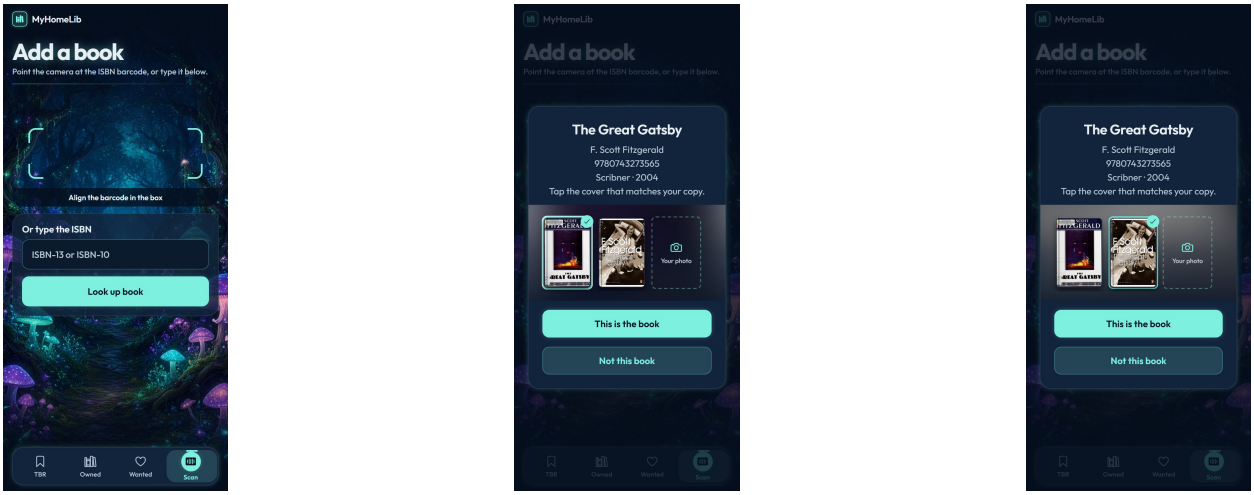


Figure 29: Scan screen (left); the edition with its covers (center); choosing the second cover (right)

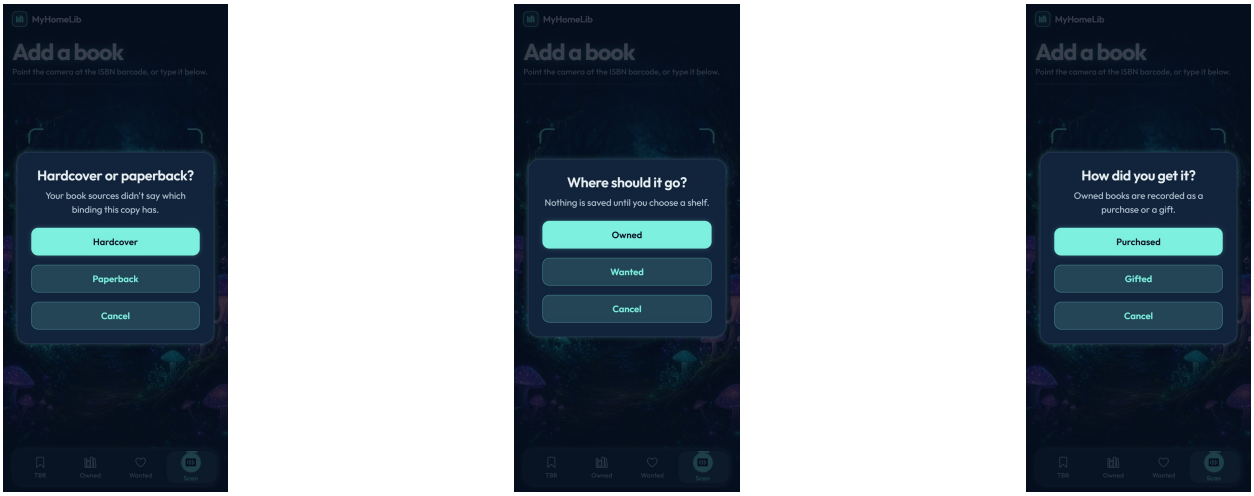


Figure 30: Binding question (left), shelf choice (center), and acquisition for Owned (right)

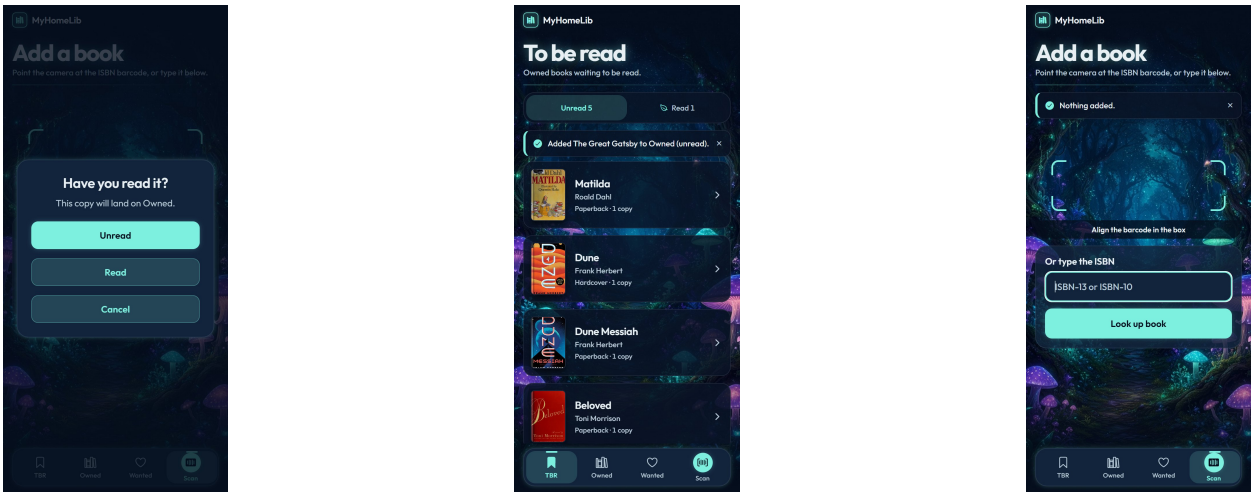


Figure 31: Reading question (left); the book is added (center); a cancelled scan adds nothing (right)

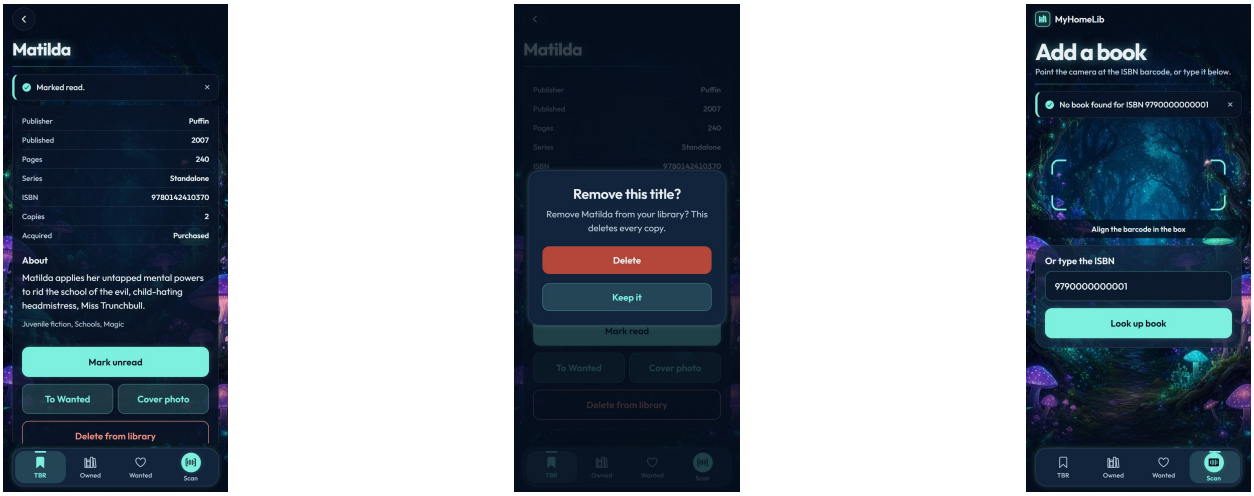


Figure 32: Marked read (left); delete confirmation (center); an unknown ISBN (right)

7.7 Sprint 7 – Hardening: Bad Input and Failures

Process	Verification	Figure
(29) We type ISBNs with a wrong check digit, the wrong length, a non-book barcode, and nothing at all. Each is refused with "… is not a valid ISBN. Check the number and try again." and no book source is asked; hyphens, spaces, an "ISBN" prefix and an ISBN-10 ending in X are accepted.	Verify ISBN validation before any network call	Table 3
(30) Resilience tests make a source time out, answer 404, 429 and 500, send broken or non-object JSON, and go offline. Each becomes one `BookSourceUnavailableError`; with Google rate-limited, Open Library still answers; with every source down the banner reads "Couldn't reach the book sources. Check your connection and try again." in red, and nothing is added. The Google key never appears in an error.	Verify graceful failure of the book sources	Table 3
(31) Persistence tests round-trip every field through JSON and a real SQLite database, store a missing acquisition as `NULL`, load a first-release database (TBR shelf, `collection` column, ISBN unique across owners), skip damaged rows, back up an unreadable save, keep two owners' copies of one ISBN apart, and store SQL in a title literally (all values are bound parameters). The same contract runs against every adapter, and a future sync adapter must pass it too.	Verify storage integrity, migration, and owner isolation	Table 3

Process	Verification	Figure
(32) The core-feature tests drive each feature through the controller, use cases, domain and a repository: an ISBN-10 and its ISBN-13 find the same merged edition; the picked cover or photo is saved; Owned needs a purchase or a named gift; a second scan of the same book, written either way, adds a copy; TBR, Wanted and the three catalog groupings list the right books; read, move, cover and delete work from a book's page; and two accounts sharing one store never see each other's books.	Verify the four core features end to end	Table 3

7.8 Sprint 8 – Accounts and Library Sync

Process	Verification	Figure
(33) Sync tests run two phones (two stores with their own clocks) against one backup: books added on each phone end up on both; the later change to a book wins; a book deleted on one phone is deleted on the other and does not come back; offline, the phone keeps working and the next sync catches up; the first sign-in moves the phone's books into the account and adds the copies of a book in both; one account never receives another's books.	Verify sync rules and owner isolation	Table 3

Process	Verification	Figure
(34) View-model tests drive the Account tab with the library: signed out, every shelf is empty and a scan is refused; a wrong password or a malformed email is explained and nobody is signed in; signing out or an expired session empties the shelves; a book added on one phone appears on the other with no tap (auto sync); with no account service, the tab says so.	Verify the Account tab and the sign-in lock	Table 3
(35) We run the web build against the shared Supabase project: the Account tab shows the sign-in form with no setup, an unreachable server gives "Couldn't reach the account service", and a signed-in browser reports "Your library is up to date". Known issue (outstanding): a delete made in the browser reached the other device only after *Sync now*. On two phones in Expo Go, signed in to one account, creating an account, adding, deleting and syncing books all worked (developer walkthrough, November 7, 2026).	Verify the app against the real service	Web build

8 References

- Anthropic. (2026). *Claude* (Opus 5.5 model) [Large language model], accessed through *Claude Code* [Computer software]. Retrieved from https://www.anthropic.com/claude-code
- Brown, S. (n.d.). *The C4 model for visualising software architecture*. Retrieved from https://c4model.com/
- Expo. (n.d.). *Expo documentation*. Retrieved from https://docs.expo.dev/
- Gamma, E., Helm, R., Johnson, R., & Vlissides, J. (1994). *Design patterns: Elements of reusable object-oriented software*. Addison-Wesley.
- Google. (n.d.). *Books API*. Retrieved from https://developers.google.com/books

- Institute of Electrical and Electronics Engineers. (2024). *IEEE standard for system, software, and hardware verification and validation* (IEEE Std 1012-2024). IEEE.
- Internet Archive. (n.d.). *Open Library APIs*. Retrieved from https://openlibrary.org/developers/api
- Martin, R. C. (2017). *Clean architecture: A craftsman's guide to software structure and design*. Prentice Hall.
- Nygard, M. (2011). *Documenting architecture decisions*. Retrieved from https://cognitect.com/blog/2011/11/15/documenting-architecture-decisions
- Sommerville, I. (2021). *Software engineering* (10th ed.). Pearson.
- Supabase. (n.d.). *Supabase documentation*. Retrieved from https://supabase.com/docs
- World Wide Web Consortium. (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*. Retrieved from https://www.w3.org/TR/WCAG22/